

Supplementary Materials

Computational design and candidate prioritization of putatively pH-sensitive EGFR binders

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S1 Potts proxy weighting and interpretation

These equations define the probability-weighted averaging of saved Potts scores after structure prediction. PROPKA supplies pKa inputs; it does not calculate these Potts expectations. A microstate is one complete combination of protonated/deprotonated states at the DD1 residues selected for protonation changes. $s_i = 1$ means site i is protonated; f_i is its protonation probability. π is the probability of the whole pattern under the assumption that sites change protonation independently. $\bar{S}$ is the arithmetic score average across patterns, and Δ_{proxy} compares the two pH endpoints. Lower Potts scores are favorable [13]. The subtraction order is neutral minus acidic: positive differences therefore indicate a lower score at pH 6.5, whereas negative differences indicate a lower score at pH 7.4. For hypothetical scores of 10 and 8 at pH 7.4 and 6.5, respectively, the difference is +2. This convention is distinct from a measured K_d : stronger binding at pH 6.5 would mean $K_d(6.5) < K_d(7.4)$. These equations do not predict either K_d or a binding free energy. Using the bound-complex PROPKA estimates [2],

$$f_i(\text{pH}) = \frac{1}{1 + 10^{\text{pH} - \text{p}K_{a,i}}}, \quad (\text{S1})$$

$$\pi(\mathbf{s}; \text{pH}) = \prod_i f_i(\text{pH})^{s_i} [1 - f_i(\text{pH})]^{1-s_i}, \quad (\text{S2})$$

$$\bar{S}(\text{pH}) = \sum_{\mathbf{s}} \pi(\mathbf{s}; \text{pH}) S_{\text{complex} - \text{minus} - \text{partners}}(\mathbf{s}), \quad (\text{S3})$$

$$\Delta_{\text{proxy}} = \bar{S}(7.4) - \bar{S}(6.5). \quad (\text{S4})$$

Scoring enumerated the binary states at DD1 residues selected for protonation changes, holding geometries and other states as configured. This arithmetic average is a model score. It does not calculate a thermodynamic partition function or the distribution of protonation states in the free proteins. Common shifts of -1 and $+1$ were applied to the center pKa estimates with the scoring setup otherwise fixed. Sensitivity counts required a strictly positive difference in both species, for the same structurally selected 30 variants. DD1_062's original generation comparison used 6.5/7.4; the initial shared post-prediction analysis used 6.5/7.3. Local reweighting of the retained complete microstate tables and unchanged pKa estimates supplied the 6.5/7.4 results reported here, after exact reproduction of all archived 6.5/7.3 endpoint scores and shift differences. No new model inference or pKa prediction was needed. The original analysis and the separately saved endpoint correction remain available in the project records.

S2 Construct identity and molecular chemistry

The assessed fusion was D46_130_model_1, derived from parent design_DD1_Human_ASP_pair_11_model_1_3_model_2. The main text calls this fusion D46; section S8 gives the short-name key. There was no additional linker or tag in the 91-residue prediction chain. The exact fusion and human D3 sequences were:

```
DD2-DD1 fusion, chain A (91 aa):
AAAKEHTVGEPLAKSEVLTLTGEGDGDSSIIIVAVVSEDNTTLKATIISGPPVLVG DYLRD
TISNSVYQIISDNNNDIILPGDVVTLKKID

Human EGFR D3, chain B (192 aa):
RKVCNGIGIGEFKDSLSINATNIKHFKNCTSI SGDLHILPVAFRGDSFTHTPPLDPQELD
ILKTVKEITGFLLIQAWPENRTDLHAFENLEIIRGR TKHQHGFSLAVVSLNITSLGLRSL
KEISDGDVVIISGNKNLCYANTINWKKLFGTSGQKTKIISNRGENSKATGQVCHALCSPE
GCWGPEPRDCVS
```

Sequence line breaks are formatting only. DD2 occupies A1–14 and DD1 A15–91. DD1 E2/D46 correspond to fusion E16/D60; the original DD1 Asp motif D11/D13 corresponds to fusion D25/D27. EGFR D3 H37/H100 maps to recorded construct positions H346/H409. Residue numbers follow the saved sequences and campaign numbering maps; we did not introduce new offsets or isoform assignments.

In the submitted AFSample3 input, glycan chain D contained two NAG, one BMA and five MAN residues; chain E contained two NAG; chains F and G each contained one NAG. They attached to receptor N19/N111/N28/N80,

respectively (recorded full positions N328/N420/N337/N389), with all 12 encoded attachment/glycosidic bonds retained. Four glycans therefore denote four entities totaling 12 saccharides. Glycan coordinates were not fixed by the protein template. Selected rational-mutant inputs retained human 6ARU-derived chemistry and modeled mouse occupancy/topology at conserved sequons. Some glycan linkage distances in those predictions, before any structural relaxation, were 0.928–1.916 Å; structures were preserved without geometric correction, and protein-only scoring did not certify these glycan contacts.

S3 Expert-guided rational substitutions

Mutation labels give the original amino acid, its DD1 position and the replacement. For example, A73R replaces alanine at position 73 with arginine; a plus sign joins substitutions made in the same sequence. This branch employed expert redesign and used DD1-local numbering: DD1_062 A73R, A76Q, I88Q and their pairwise/triple combinations (seven variants); DD1_195 A7Q, A7R, F43Y, Q22Y, all four A7-plus-aromatic pairs, Q22Y+F43Y, and the two A7-plus-Q22Y+F43Y combinations (eleven variants); and DD1_310 L18S, L18T, L18S+A32S and L18T+A32S (four variants). The saved sequence records establish exact identities independently of the shorthand mutation labels.

S4 Supporting campaign assessments

S4.1 Auxiliary separate-chain DD2 branch

The separate-chain branch combined Foundry RFD3, soluble-weight ProteinMPNN [7] and RF3 [6]. DD2 was designed against fixed DD1_195 or DD1_310, with ten sequences per backbone, target redesign disabled, a 5 Å interface exclusion, and RF3 inference with 50 diffusion steps and one sample. The retained analysis comprised 1,040 complexes per parent and selected the top 5% by ipSAE separately. These predictions contained no EGFR and were not pH-conditioned.

S4.2 Rational-mutant assessment details

Seven DD1_062, eleven DD1_195 and four DD1_310 variants produced 220 Boltz-2 samples [1]. Selecting the highest model-confidence score gave 44 species-specific models, all sample 0 after comparison of the five samples. The substitutions introduced charged/polar residues at DD1_062 A73/A76/I88 and DD1_195 A7, tyrosines at DD1_195 Q22/F43, and hydroxyl-bearing residues at DD1_310 L18/A32 (Supplementary section S3). These describe the chemical perturbations; the retained records do not establish a contact-based structural rationale for selecting each site.

DockQ relative to the original RFD3 interface was 0.0169–0.0700 (median 0.0221), indicating poor design-pose retention in the selected models. Agreement with same-species unmutated Boltz-2 references was 0.0441–0.7706 (median 0.0874). Historical template/glycan settings differed between parent and mutant predictions, so these comparisons do not isolate a mutation’s causal effect. JustHISpKa completed 87 of 88 unforced sites. Mouse DD1_062 A76Q+I88Q H100 required conditional neighboring-acid states, giving 6.771 when deprotonated and 4.716 when protonated. All 44 protein-only DeltaForge derivatives [5] returned predictions (33.42–645.11 nM).

S4.3 APNovo_0293 disulfide-assignment control

Preparation chemistry was also assessed for APNovo_0293: C2–C84 alone versus C2–C84 plus C10–C73 on fixed coordinates changed the four JustHISpKa [8] estimates by at most 0.006 (Table S1). This comparison concerns specified computational chemistry, without determining the expressed disulfide state.

Table S1: JustHISpKa predictions for APNovo_0293 under two binder-disulfide assignments. H37/H100 denote EGFR D3 sites.

Binder assignment	Human H37	Human H100	Mouse H37	Mouse H100
C2-C84	3.463	5.932	4.434	6.492
C2-C84 and C10-C73	3.469	5.934	4.437	6.492

Table S2: Boltz-2 confidence scores for D46 fusion and its DD1 parent. Each column is one retained prediction; no pH parameter was supplied.

Metric	Fusion human	Fusion mouse	DD1 human	DD1 mouse
pTM	0.904828	0.928673	0.937278	0.943511
ipTM	0.757544	0.836490	0.840341	0.858445
ipSAE	0.468054	0.697520	0.638875	0.661752
Interface pLDDT (0-1)	0.837847	0.825329	0.878838	0.893332

S4.4 D46 species-complex confidence and ionization annotations

JustHISpKa [8] predicted human H37/H100 of 4.400/7.029 for the fusion and 4.964/7.593 for DD1. Mouse H37 was 4.333 for the fusion and 5.477 for DD1; parent mouse H100 was 6.985. Fusion mouse H100 had no unique estimate without imposing neighboring-acid states: fixing binder E16/D60 as deprotonated gave 8.173, whereas fixing them as protonated gave 2.338. Those acidic residues lay approximately 3.83/3.14 Å from the histidine ring nitrogens. The values refer to imposed protonation patterns and were not averaged. Independently folded DeltaForge sequence requests via LigandAI [5] predicted fusion human/mouse K_d values of 34.54/130.35 nM and parent values of 63.79/102.33 nM; those service structures were distinct from the Boltz-2 models.

S5 Recovered AFSample3 projection and candidate inventory

The archived workflow contains 1,001 entries: 1,000 seed/sample predictions plus a run-level model entry. AF-Sample3/AlphaFold 3 supplied these predictions [3, 4]. The four cluster counts include that extra entry and are not populations of 1,000 independent samples. The clustering operated on shared chain-A $C\alpha$ distance matrices, using UMAP [10] (15 neighbors; minimum distance 0.1) and a 15-neighbor connectivity-constrained Ward hierarchy implemented with scikit-learn [9]. Four clusters maximized the archived silhouette criterion (0.53793). The plot uses the saved two-dimensional coordinates and cluster assignments; neither was recalculated. The original PNG and the matching CSV (verified against the archived hash) are preserved alongside this draft.

The full inventory contains 394 distinct sequences with complete paired native ipSAE, Potts scores and DeltaForge affinity predictions. Main Figure 3 and the species-specific affinity plot (Figure S8) use this same inventory, including all 100 selected Novo redesigns and 100 Foundry candidates. Two candidates lack a unique unforced mouse His100 estimate and retain two separate neighboring-acid-state predictions each in the pKa plots. Conditional values are never averaged or substituted into unforced fields. Scores join exact full sequences and species-specific native model identities.

S6 Candidate histidine pKa map

The current main and full histidine-pKa maps use all 394 selected sequences: the original 77 constructs, 17 rational additions, 100 LigandForge redesigns, 100 original Foundry designs and 100 Novo redesigns. All have complete paired confidence, affinity and Potts scores; human receptor pairs are unforced for all 394, and mouse pairs are unforced for 392 with the two conditional cases shown separately. Current plotted receptor estimates use source-linked bound-complex JustHISpKa [8], including the completed 52-site successor for the ten original LigandForge binders and three DD1 parents. Tables S10, S7 and S12 retain cohort identities and values. Table S3 preserves the original 77-candidate identity lookup and historical estimates unchanged: J denotes JustHISpKa on a bound complex, P denotes PROPKA on a bound complex, and R denotes JustHISpKa on receptor-only

coordinates. The historical LigandForge entries retain PROPKA 3.5.1 and preparation warnings [2, 5]; R entries are receptor-only controls. These historical annotations are distinct from the current bound-complex successors and were not substituted for unavailable predictions. Bold table values satisfy the inclusive 6.5–7.4 criterion independently of species. Differences between predictors or preparation scopes do not establish calibrated binder-induced pKa shifts. APNovo_0293 uses the existing recovered unforced calculations with specified C2–C84 and C10–C73 binder disulfides. The alternative assignment in Table S1 leaves window membership unchanged. Unforced mouse H100 values in DD1_062 A76Q+I88Q and D46 fusion remain blank. Their separate neighboring-acid-state calculations are now displayed with explicit labels in the main and full supplementary pKa maps. The former qualifies through H37; membership of the latter is unresolved. Threshold membership uses unrounded stored values. Human and mouse pKas were not averaged. The original lookup table is retained unchanged; current plots incorporate the completed source-linked JustHISpKa successors. Points were not displaced to avoid overlaps. Display names APNovo_n expand to DD1_Human_EGFR_APNovo_n; LF_n expands to Ligand_AI_b6a74871_Design_n; DD1_parent/suffix expands to DD1_parent_phswitch_suffix. Other names retain construct or expert-redesign identities. ID is the stable 77-candidate point identifier. The CSV linked to the original calculations retains exact sequences, hashes, source tables, prediction scope and calculation status.

Table S3: Retained H37/H100 predicted pKa values by candidate and species. Bold: in 6.5–7.4; –: no unforced estimate.

ID	Candidate	Scope	Human H37	H100	Mouse H37	H100
1	APNovo_0026	J	6.690	5.422	7.189	5.789
2	APNovo_0037	J	4.576	6.103	4.625	6.547
3	APNovo_0293	J	3.469	5.934	4.437	6.492
4	APNovo_0394	J	4.308	6.415	3.123	6.292
5	APNovo_0417	J	3.357	5.918	3.662	5.347
6	APNovo_0470	J	4.616	6.815	4.461	6.853
7	APNovo_0599	J	4.211	5.686	4.353	6.105
8	APNovo_0754	J	7.364	6.989	7.514	7.030
9	APNovo_0860	J	4.800	4.318	5.355	4.632
10	APNovo_0960	J	4.545	6.991	3.828	5.457
11	D46_130 DD1 reference	J	4.964	7.593	5.477	6.985
12	D46_130 fusion	J	4.400	7.029	4.333	–
13	DD1_062/0312	J	2.936	3.516	3.723	5.804
14	DD1_062/0333	J	4.521	6.854	4.018	6.715
15	DD1_062/0436	J	4.397	4.452	4.271	4.363
16	DD1_062/0514	J	4.385	6.783	3.976	4.998
17	DD1_062/0557	J	4.078	5.092	4.512	5.424
18	DD1_062/0629	J	4.832	7.464	4.677	5.102
19	DD1_062/0765	J	4.351	6.480	4.390	6.732
20	DD1_062/0792	J	4.980	3.251	4.925	6.991
21	DD1_062/0861	J	4.048	6.775	4.374	6.541
22	DD1_062/0955	J	3.071	6.550	3.484	6.598
23	DD1_195/0043	J	6.896	4.887	6.894	5.336
24	DD1_195/0056	J	5.569	0.423	6.970	6.739
25	DD1_195/0096	J	5.497	3.938	5.182	4.268
26	DD1_195/0236	J	6.988	7.083	7.173	4.646
27	DD1_195/0304	J	5.406	6.890	3.805	6.570
28	DD1_195/0458	J	6.740	5.073	5.207	6.479
29	DD1_195/0550	J	6.304	6.749	2.435	7.057
30	DD1_195/0591	J	6.187	1.456	6.900	4.812
31	DD1_195/0757	J	6.352	5.761	8.388	6.616
32	DD1_195/0759	J	6.989	4.467	6.330	4.403
33	DD1_310/0217	J	5.488	5.066	5.484	6.178
34	DD1_310/0268	J	3.917	5.014	4.863	4.607
35	DD1_310/0412	J	5.471	5.685	6.457	5.854
36	DD1_310/0504	J	4.724	4.974	4.453	5.544
37	DD1_310/0516	J	4.472	6.354	4.880	5.559
38	DD1_310/0616	J	4.920	5.016	4.417	5.086
39	DD1_310/0719	J	4.095	5.522	4.582	5.283
40	DD1_310/0735	J	4.272	4.527	5.583	6.032
41	DD1_310/0935	J	3.581	5.691	4.128	6.812
42	DD1_310/0946	J	4.304	6.693	4.812	6.626
43	DD1_062	R	6.993	6.936	6.793	6.529
44	DD1_195	R	6.944	6.155	6.866	6.190
45	DD1_310	R	6.886	6.871	7.133	6.900
46	LF_507	P	3.976	4.736	2.001	6.306
47	LF_559	P	3.060	5.963	3.605	5.320
48	LF_671	P	4.811	5.461	5.052	5.262
49	LF_698	P	3.460	6.124	3.376	6.753
50	LF_713	P	3.129	3.889	3.575	5.587
51	LF_774	P	6.556	4.588	6.058	5.758
52	LF_927	P	3.490	4.238	3.595	4.614
53	LF_961	P	3.471	4.777	3.791	6.490
54	LF_971	P	3.330	5.243	3.902	5.416

Table S4 (continued)

ID	Candidate	Scope	Human H37	H100	Mouse H37	H100
55	LF_973	P	3.849	4.895	6.385	5.395
56	DD1_062_A73R	J	6.374	6.594	6.637	6.962
57	DD1_062_A73R_A76Q	J	7.114	6.533	3.458	6.928
58	DD1_062_A73R_A76Q_I88Q	J	1.141	5.353	4.921	6.429
59	DD1_062_A73R_I88Q	J	3.585	5.933	5.419	6.976
60	DD1_062_A76Q	J	6.514	6.893	6.783	6.508
61	DD1_062_A76Q_I88Q	J	6.674	7.114	6.738	–
62	DD1_062_I88Q	J	6.563	6.382	2.645	5.772
63	DD1_195_A7Q	J	6.966	5.511	6.311	7.426
64	DD1_195_A7Q_F43Y	J	6.859	4.612	6.931	5.309
65	DD1_195_A7Q_Q22Y	J	5.635	4.673	6.567	4.984
66	DD1_195_A7Q_Q22Y_F43Y	J	4.648	6.649	6.979	5.120
67	DD1_195_A7R	J	6.395	4.771	6.653	5.396
68	DD1_195_A7R_F43Y	J	6.814	3.634	4.519	5.935
69	DD1_195_A7R_Q22Y	J	6.671	5.721	6.454	4.757
70	DD1_195_A7R_Q22Y_F43Y	J	6.169	6.137	4.722	6.424
71	DD1_195_F43Y	J	4.285	6.562	5.617	6.282
72	DD1_195_Q22Y	J	6.145	4.515	6.701	6.525
73	DD1_195_Q22Y_F43Y	J	6.681	7.074	5.009	5.778
74	DD1_310_L18S	J	6.497	6.222	4.505	5.411
75	DD1_310_L18S_A32S	J	6.417	5.394	5.170	6.772
76	DD1_310_L18T	J	6.871	6.894	3.102	5.094
77	DD1_310_L18T_A32S	J	6.512	6.313	7.056	6.335

S7 Archived and reweighted Potts endpoint comparison

Table S4 supplies the endpoint comparison for all five rows of Table S5, in the same order. The archived comparison was pH 6.5/7.3. Its retained complete scores for all considered protonation patterns, bound-complex PROPKA pKa estimates, geometries, other protonation states and selected sequence identities were held fixed while Henderson–Hasselbalch weights were recomputed at pH 7.4. No new Potts model evaluation, structure prediction or pKa calculation was performed.

Table S4: Archived and reweighted proxy differences at the original pKa estimates (zero common shift), in model units. $\Delta_{7.3} = \bar{S}(7.3) - \bar{S}(6.5)$ is the archived comparison; $\Delta_{7.4} = \bar{S}(7.4) - \bar{S}(6.5)$ uses recomputed weights on the same microstate scores. Positive differences indicate a lower weighted score at pH 6.5. Values are rounded to six decimals for display; exact reproduction was checked against the unrounded archived records. These differences are not measured affinities or binding free energies.

Parent / variant suffix	Human EGFR complex		Mouse EGFR complex	
	$\Delta_{7.3}$	$\Delta_{7.4}$	$\Delta_{7.3}$	$\Delta_{7.4}$
DD1_062 / 0955	0.811236	0.861509	1.301748	1.397236
DD1_062 / 0629	0.165510	0.171079	3.483351	3.716210
DD1_062 / 0436	0.247944	0.277372	0.573952	0.654523
DD1_195 / 0591	-0.064084	-0.066831	-1.041998	-1.097423
DD1_310 / 0268	-0.196397	-0.205550	-0.217680	-0.227800

Across all 60 retained complexes and common pKa shifts of -1 , 0 and $+1$, the archived pH 6.5/7.3 endpoint scores and shift differences were reproduced exactly at the stored numerical precision (maximum absolute error 0.0). Changing the probability weights changed no signs in the 180 species/shift comparisons; the largest absolute change in the zero-shift difference across 60 complexes was 0.232859 model units. The same three DD1_062 variants remained positive in both species under all three shifts at both endpoints. These checks validate consistency of the endpoint reweighting within the independent-site weighting model, rather than experimental pH selectivity.

Table S5 combines the structural confidence scores and reweighted pH differences for the three variants that retained the acidic direction under every tested pKa shift, together with the leading structural selections from the other two starting binders.

Table S5: Three DD1-062 variants that retain the desired trend under pKa shifts and the leading structural selections from the other parents. The first three rows retain positive model-score differences in both species at all three tested common pKa shifts. Differences are Henderson–Hasselbalch reweighted model scores at pH 7.4 minus 6.5, reusing the saved scores for each protonation pattern and original pKa estimates (Table S4). The human and mouse differences are labeled Δ_H and Δ_M . A positive value denotes a lower average score at pH 6.5 and does not quantify a binding advantage.

Variant	Human ipSAE	Mouse ipSAE	Mean	Δ_H	Δ_M
062/0955	0.898849	0.868393	0.883621	0.862	1.397
062/0629	0.870402	0.840174	0.855288	0.171	3.716
062/0436	0.848198	0.814891	0.831545	0.277	0.655
195/0591	0.862079	0.917434	0.889757	-0.067	-1.097
310/0268	0.812539	0.841756	0.827147	-0.206	-0.228

S8 Short names and exact saved identifiers

Short names make the main text easier to read without changing any sequence or original record. In names such as 062/0955, 062 identifies the starting DD1 binder and 0955 identifies its redesigned sequence. The following key gives the exact saved names for designs discussed individually in the main text. Table S4 retains its original display names and point identifiers so that earlier figures and source tables can still be matched.

DD1-062 DD1_062
DD1-195 DD1_195
DD1-310 DD1_310
062/0955 DD1_062_phswitch_0955
062/0629 DD1_062_phswitch_0629
062/0436 DD1_062_phswitch_0436
195/0591 DD1_195_phswitch_0591
310/0268 DD1_310_phswitch_0268
D46 D46_130_model_1
Novo-0293 DD1_Human_EGFR_APNovo_0293
Novo-0037 DD1_Human_EGFR_APNovo_0037
Novo-0470 DD1_Human_EGFR_APNovo_0470

The unchanged DD1 sequence within D46 came from `design_DD1_Human_ASP_pair_11_model_1_3_model_2`. Novo-*n* denotes an AlphaProtein Novo design originally saved as `DD1_Human_EGFR_APNovo_n`. LigandForge’s LF-*n* names denote `Ligand_AI_b6a74871_Design_n`.

The LigandForge web-summary PDF identifies the completed generation:

b6a74871-d6d7-4e4f-9e54-07c5cf418916

Matching its H-Q-T-H pattern to the design fragment, with successive residue offsets of 38, 22 and 3, established the mapping in Table S3. The web labels, local fragment positions and project reference positions are distinct numbering systems.

S9 Technical prediction and scoring settings

These details support reproduction of the calculations. A random seed initializes a sampling run. Recycles are repeated passes through a structure-prediction model. A multiple sequence alignment (MSA) supplies aligned related sequences; a template supplies an existing structure. A checkpoint is the saved set of trained model parameters. These terms describe software settings, not separate biological experiments.

S9.1 Boltz-2 settings

Boltz-2 [1] predictions shared three recycles, 200 sampling steps, cached receptor MSAs and empty binder MSAs. One diffusion sample was used for discovery/redesign and five for rational mutants. Seeds were 101 for motif-directed DD1/DD2, LigandForge and rational mutants, and 0 for AlphaProtein Novo and Proton-PottsMPNN. Motif-directed DD1/DD2 and LigandForge used no templates; AlphaProtein Novo and Proton-PottsMPNN used target-only templates. Rational mutants used an unforced receptor-only 6ARU template, including homologous mouse inputs, enabled potentials, full PAE and one maximum parallel sample. Chemistry differed by campaign as described below.

S9.2 HBPLUS features and protonation labeling

A hydrogen-bond feature records the geometry of a potential donor-acceptor contact. Proton-PottsMPNN extends PottsMPNN [13, 12]. For this redesign, separate HBPLUS analyses were combined as described below. The starting designs came from Foundry [6]; the redesign implementation was run separately. The FLAML-based labeling step used these structural features to classify protonation states for the design model, rather than assigning a numerical pH.

The v6 preparation pipeline removed input hydrogens and invoked HBPLUS [11] with H-acceptor/donor-acceptor cutoffs 3.2/4.0 Å and default 90° minimum angular criteria. Default and Asp/Glu carboxyl-oxygen-donor passes

were merged and deduplicated. Histidine features used default-pass bonds; acid features used the merged pool. Those geometric features supplied the FLAML structural protonation labeller and checkpoint context, without defining a numerical pH. Post-prediction Potts scoring used the same context preparation.

S9.3 AFSample3 settings and clustering

AFSample3 used seeds 1–200, five samples per seed, ten recycles, an MSA cap of 1,024, a random fraction of 0.4 and MSA shuffling. Binder MSAs and templates were empty. EGFR used a matched, previously calculated MSA and a protein-coordinate medoid template: a representative structure chosen by proximity to other structures. The input included four covalently attached glycans and no explicit pH condition.

Pairwise backbone distances describe each structure. PCA, UMAP and t-SNE reduce these many distances to a lower-dimensional display. Ward clustering groups similar points hierarchically; the silhouette score evaluates separation between groups. A medoid is an actual representative structure near the center of its group. These geometric tools do not establish functional states or their populations.

Shared chain-A C α distance matrices for 1,001 entries (1,000 predictions plus a run-level structure) supplied PCA, UMAP [10] (15 neighbors; minimum distance 0.1) and scikit-learn t-SNE [9] (perplexity 30; seed 42) visualizations. A 15-neighbor connectivity-constrained Ward hierarchy on UMAP selected four clusters by maximizing Euclidean silhouette over 2–20 clusters. Five highest-pTM models per cluster were aligned to its full-cluster C α RMSD medoid. Supplementary Figure S1 shows the recovered UMAP projection, four retained cluster labels, medoids and selected Figure 1 examples. Figure S2 shows the four actual medoid models. The original workflow applied no contact/occlusion state classifier or population estimator; the subsequent display comparison is described in section S10. The retained run-root ranking CSV contained only five samples from seed 200; its leading sample had global ipTM 0.67 but fusion–EGFR pair ipTM 0.24. That file was not an ensemble-wide ranking.

S10 AFSample3 medoids and Figure 1 example selection

A medoid is an actual model that best represents its cluster under the selected distance measure, rather than an averaged structure. The archived analysis selected each medoid using whole-fusion C α RMSD to its cluster members. These medoids need not be closest to the center of the two-dimensional UMAP picture. Table S6 identifies them; Figure S2 displays fusion chain A after DD1 alignment to the matched human-bound parent, with one camera and scale for all four models. Receptor and glycan chains are omitted from these display copies, with native predictions preserved.

Table S6: Archived AFSample3 cluster medoids. Seed/sample identifies a native prediction of the D46 construct defined in section S2. Cluster 2 includes the additional run-level entry; the prediction counts sum to 1,000. Mean RMSD refers to the archived whole-fusion C α comparison to cluster members, not the later DD1-only alignment.

Cluster	Entries	Predictions	Seed/sample	Mean C α RMSD to members (Å)
1	378	378	166/3	3.660
2	345	344	121/3	4.632
3	198	198	83/4	2.946
4	80	80	176/4	2.380

The later Figure 1 display selection checked the exact fusion and receptor sequences in all 1,000 seed/sample models. DD1 residues 15–91 of each fusion were aligned to the independently predicted, sequence-matched 77-residue DD1-only human complex. That reference defines sixteen DD1 residues with a heavy atom within 4.5 Å of human EGFR. A DD2–DD1 proximity was counted at the same cutoff, excluding atom pairs separated by four or fewer residues along the fusion sequence. This excludes proximity caused only by the immediate covalent tether. The receptor in the reference complex supplies a projected target placement; it is not the native AFSample3 docking orientation.

Illustrative examples were selected from models with DD1 C α alignment RMSD below 1.5 Å. Main Figure 1C

(cluster 3, seed 195/sample 3) illustrates the intended exposed form at pH 6.5. It has a 0.99 Å DD1 RMSD, no reference interface residues near DD2, and a minimum DD2-projected-receptor distance of 7.31 Å. Main Figure 1D (cluster 4, seed 149/sample 2) illustrates the intended covered form at pH 7.4. It has a 1.10 Å DD1 RMSD, eight reference interface residues near DD2, and a 2.92 Å separation between DD2 His6 ND1 and DD1 Glu16 OE2; DD2 would overlap the reference receptor placement. These two examples are not medoids and do not label their entire clusters. The pH labels describe the intended mechanism, not states established by pH-conditioned predictions. Their fusion-chain pTM values are 0.67/0.70 and fusion-EGFR chain-pair ipTM values are 0.25/0.21 for C/D, respectively. The higher global ipTM scores include EGFR-glycan interfaces and do not establish confidence in the fusion-target interaction.

This inspection identifies possible DD2 placements relative to DD1. It does not estimate equilibrium populations, infer a pH-dependent transition, establish unfolding, or measure binding. The original DD2 histidine-lysine distance comparison and these additional histidine-carboxylate proximities assess different contacts. All source files, exact model identifiers, saved projection values and display transforms remain linked to the local source records.

S11 Completed LigandForge-derived redesign predictions

The cohort contains 100 unique sequences, ten per LigandForge parent, selected from verified completed pairs by mean human/mouse ipSAE, then weaker-species score and stable design identity. LigandAI is the platform, LigandForge the original generator, Proton-PottsMPNN the standalone redesign implementation and DeltaForge the affinity scorer [12, 13, 5, 6]. Foundry is related-framework attribution, not the execution framework for this redesign.

The completed run was aggregated at 2026-10-07T03:34:09-04:00 (EDT) on 7 October 2026. All 400 requested unforced JustHISpKa estimates and 200 native-complex DeltaForge affinity predictions passed source-linked integrity checks. All 100 redesigns have complete H37/H100 pairs per species; 66 human and 61 mouse pairs meet the inclusive 6.5–7.4 criterion. Native protein-only coordinates were preserved without minimization. AmberTools 26 ff14SB and NRC correction were used with JustHISpKa [8]; each result maps receptor chain B H37/H100 to full EGFR H346/H409 and its actual global site identifier. No PROPKA or conditional substitution was used. Separate optional PROPKA coverage flags do not invalidate independently verified completed JustHISpKa outputs.

The successor retained four receptor disulfides in every complex and eight additional verified intra-binder bonds across five designs. LF-971/0171 C37–C52, LF-671/0599 C2–C7 and LF-559/0061 C3–C16 were preserved in both species; LF-961/0677 C15–C52 and LF-713/0037 C2–C42 were preserved only in Human. No corresponding Mouse bond was imposed. Mouse LF-559/0061 has a short 1.589 Å SG–SG geometry warning; its source coordinates and estimates are retained. The remaining 192 complex records were reused byte-for-byte.

DeltaForge used native protein-only PDBs (receptor B, binder A), SDK 0.6.4, deployed scorer `v10_2_unified_parallel_2026_05_08` and affinity calibration `v10_boltz2_gbr_2026_03_29` [5]. Both served binding ΔG (kcal/mol) and K_d (nM) are reported without conversion between them. These predictions and Boltz-2 structures have no pH input; they do not predict affinity differences between pH 6.5 and 7.4 or folding free energies.

Short names LF-*p/n* expand exactly to `LigandAI_p_phswitch_n`. The frozen selection order, sequences, parent links, model hashes, raw-output hashes and deployed versions remain in the completed run’s source tables. Table S7 follows that order. Bold pKas are in 6.5–7.4; all sites completed. The affinity columns give served predictions, rounded only for display.

Table S7: Completed receptor-site pKa and binding-affinity predictions for all 100 LigandForge-derived redesigns. H/M denote Human/Mouse; K_d is nM and ΔG is kcal/mol.

Design	H H37	H H100	M H37	M H100	H K_d	M K_d	H ΔG	M ΔG
LF-698/0123	4.042	6.870	5.624	6.939	1.09	51.49	-12.23	-9.94
LF-698/0088	3.907	6.111	4.857	6.586	107.05	111.42	-9.51	-9.49
LF-698/0641	4.534	6.668	4.197	6.902	252.03	163.05	-9.00	-9.26
LF-698/0934	4.912	6.999	3.190	6.382	102.00	88.61	-9.54	-9.62

Table S7: Completed LigandForge-derived redesign predictions (continued).

Design	H H37	H H100	M H37	M H100	H K_d	M K_d	H ΔG	M ΔG
LF-698/0809	7.197	6.663	5.985	7.196	43.43	64.88	-10.04	-9.81
LF-698/0786	2.016	7.056	4.888	4.981	40.00	52.75	-10.09	-9.93
LF-698/0284	6.566	6.325	4.220	6.937	202.50	94.26	-9.13	-9.58
LF-698/0068	0.343	6.146	1.353	6.235	62.26	8.51	-9.83	-11.01
LF-698/0507	6.712	6.879	5.713	6.677	48.11	20.39	-9.98	-10.49
LF-698/0923	3.986	6.189	6.596	6.678	54.39	74.65	-9.91	-9.72
LF-971/0281	6.510	6.636	4.878	7.145	86.42	45.74	-9.64	-10.01
LF-971/0786	6.479	0.649	3.666	5.909	84.25	40.74	-9.65	-10.08
LF-971/0677	5.301	5.836	5.426	6.126	95.70	68.79	-9.58	-9.77
LF-971/0767	6.622	7.249	7.252	7.187	50.59	14.01	-9.95	-10.71
LF-971/0945	4.267	7.916	5.067	7.938	658.05	819.55	-8.43	-8.30
LF-971/0171	3.963	6.613	7.274	3.707	76.89	94.46	-9.71	-9.58
LF-971/0551	7.138	4.195	4.404	7.071	114.49	613.34	-9.47	-8.48
LF-971/0128	6.223	4.934	4.122	6.024	85.37	0.90	-9.64	-12.34
LF-971/0842	6.479	5.769	4.326	6.811	1.21	125.68	-12.17	-9.41
LF-971/0303	6.309	3.583	3.871	6.476	105.08	97.28	-9.52	-9.57
LF-507/0434	4.672	5.763	4.259	4.131	112.31	7.65	-9.48	-11.07
LF-507/0021	6.549	8.220	7.361	6.520	29.60	0.67	-10.27	-12.52
LF-507/0205	6.857	6.842	5.938	5.532	139.27	34.32	-9.35	-10.18
LF-507/0566	7.565	6.554	7.132	7.828	109.01	2268.59	-9.50	-7.70
LF-507/0135	5.483	7.177	6.075	6.553	17.43	26.88	-10.58	-10.33
LF-507/0807	5.459	6.624	6.878	7.030	181.09	177.30	-9.20	-9.21
LF-507/0272	3.616	6.829	6.813	4.757	3.78	132.24	-11.49	-9.38
LF-507/0089	4.320	5.472	4.785	5.018	80.67	9.25	-9.68	-10.96
LF-507/0736	4.203	4.370	4.609	5.957	99.43	318.95	-9.55	-8.86
LF-507/0801	6.224	4.061	4.784	4.722	249.79	46.67	-9.01	-10.00
LF-927/0808	4.971	6.725	4.954	6.517	1.29	1.17	-12.13	-12.19
LF-927/0987	4.347	6.103	5.988	7.101	192.55	40.06	-9.16	-10.09
LF-927/0665	6.897	2.979	7.090	6.490	26.44	0.53	-10.34	-12.65
LF-927/0492	6.736	4.992	6.958	6.874	42.65	27.52	-10.05	-10.31
LF-927/0409	6.899	7.074	6.140	4.346	264.14	83.15	-8.97	-9.66
LF-927/0209	6.750	6.910	2.999	6.708	94.65	71.64	-9.58	-9.75
LF-927/0211	6.173	1.766	4.920	2.218	68.53	2.44	-9.77	-11.75
LF-927/0892	6.797	7.924	5.005	7.227	58.56	50.14	-9.87	-9.96
LF-927/0361	3.457	5.953	5.113	1.942	3.34	39.82	-11.56	-10.10
LF-927/0831	4.726	5.887	5.217	5.443	63.15	44.95	-9.82	-10.02
LF-961/0411	5.593	6.246	7.324	8.377	84.25	188.05	-9.65	-9.18
LF-961/0506	4.999	4.840	5.813	7.054	84.18	166.35	-9.65	-9.25
LF-961/0742	4.631	7.141	5.994	7.307	105.18	185.75	-9.52	-9.18
LF-961/0776	5.314	6.114	4.248	6.646	214.64	366.15	-9.10	-8.78
LF-961/0750	4.587	3.326	4.675	2.722	5.25	24.18	-11.30	-10.39
LF-961/0826	7.009	8.035	3.932	6.432	273.23	474.23	-8.95	-8.63
LF-961/0677	5.302	7.095	5.284	7.330	87.98	17.13	-9.63	-10.60
LF-961/0475	6.424	6.819	7.654	3.744	294.15	60.21	-8.91	-9.85
LF-961/0207	4.371	7.288	4.351	7.956	204.25	84.57	-9.13	-9.65
LF-961/0695	7.101	6.544	7.407	6.195	45.55	117.38	-10.02	-9.45
LF-713/0076	8.900	4.289	2.712	1.592	242.31	131.59	-9.03	-9.39
LF-713/0558	6.331	6.798	4.588	6.897	22.18	2.20	-10.44	-11.81
LF-713/0026	3.756	7.189	6.955	5.163	14.53	36.46	-10.69	-10.15
LF-713/0760	5.077	6.812	3.833	6.385	66.91	45.12	-9.79	-10.02
LF-713/0057	4.191	6.703	6.833	7.093	92.00	128.24	-9.60	-9.40
LF-713/0443	3.291	6.995	5.090	5.878	1.53	3.40	-12.03	-11.55
LF-713/0068	7.033	7.102	6.975	5.411	152.31	25.92	-9.30	-10.35
LF-713/0853	4.778	2.551	4.450	7.197	162.61	3.29	-9.26	-11.57
LF-713/0037	4.895	6.028	4.172	7.122	115.77	0.97	-9.46	-12.29
LF-713/0157	4.600	6.689	4.698	6.932	28.42	47.24	-10.30	-9.99
LF-671/0188	6.815	5.899	6.878	6.168	204.75	175.64	-9.13	-9.22
LF-671/0434	7.042	5.310	7.109	5.560	28.16	31.45	-10.30	-10.24
LF-671/0016	6.317	6.884	6.824	5.392	86.48	72.90	-9.64	-9.74
LF-671/0721	6.612	6.865	4.846	5.726	210.97	161.52	-9.11	-9.27
LF-671/0887	7.047	6.673	6.950	6.820	30.11	28.46	-10.26	-10.29
LF-671/0395	6.682	7.052	6.659	7.029	202.93	65.32	-9.13	-9.80
LF-671/0084	6.650	5.795	6.606	4.824	140.88	122.89	-9.35	-9.43
LF-671/0730	6.410	7.166	4.055	6.416	61.35	37.86	-9.84	-10.13
LF-671/0599	6.744	5.664	7.125	4.710	129.21	139.18	-9.40	-9.35
LF-671/0042	6.871	6.316	6.844	6.457	108.70	643.20	-9.50	-8.45
LF-559/0456	4.324	5.777	5.241	7.145	63.12	104.72	-9.82	-9.52
LF-559/0770	7.149	6.887	7.124	6.941	102.71	69.19	-9.53	-9.77
LF-559/0325	5.580	7.023	2.691	7.333	88.74	0.75	-9.62	-12.45
LF-559/0335	4.886	6.716	5.232	5.964	133.23	87.64	-9.38	-9.63
LF-559/0061	6.777	6.615	6.985	6.485	235.54	64.56	-9.04	-9.81
LF-559/0003	7.019	6.592	5.321	5.884	56.65	69.20	-9.89	-9.77
LF-559/0788	5.034	6.672	8.045	6.967	91.77	55.07	-9.60	-9.90
LF-559/0336	7.030	6.565	8.110	7.343	184.39	62.43	-9.19	-9.83
LF-559/0216	5.232	6.931	-0.266	6.626	26.05	121.36	-10.35	-9.44
LF-559/0129	5.714	6.960	7.231	7.052	31.34	20.64	-10.24	-10.48
LF-973/0351	4.721	7.040	4.800	6.573	0.88	2.53	-12.36	-11.73
LF-973/0349	4.448	6.493	5.189	7.934	13.66	101.62	-10.73	-9.54
LF-973/0984	5.375	6.384	4.740	6.919	133.75	88.41	-9.38	-9.62
LF-973/0380	7.138	6.709	5.317	7.244	364.58	195.90	-8.78	-9.15

Table S7: Completed LigandForge-derived redesign predictions (continued).

Design	H H37	H H100	M H37	M H100	H K_d	M K_d	H ΔG	M ΔG
LF-973/0867	4.304	7.279	5.199	7.557	33.50	5.52	-10.20	-11.27
LF-973/0009	5.123	7.126	4.740	3.939	57.90	23.58	-9.87	-10.41
LF-973/0523	6.748	6.744	5.558	1.728	184.28	41.59	-9.19	-10.07
LF-973/0544	5.271	7.347	4.128	7.322	50.65	28.56	-9.95	-10.29
LF-973/0621	4.653	6.969	5.386	6.396	78.65	2.64	-9.69	-11.70
LF-973/0526	5.174	7.407	4.897	7.121	2.20	14.88	-11.81	-10.68
LF-774/0377	4.068	3.982	3.903	4.633	37.02	116.93	-10.14	-9.46
LF-774/0427	4.020	5.799	3.410	4.790	86.52	204.97	-9.64	-9.12
LF-774/0167	4.164	6.821	4.720	7.378	80.85	74.65	-9.68	-9.72
LF-774/0080	4.758	1.871	4.247	6.798	161.14	0.73	-9.27	-12.46
LF-774/0333	3.290	6.953	4.653	6.855	28.38	30.27	-10.30	-10.26
LF-774/0090	4.045	1.454	4.623	6.039	1.49	99.70	-12.04	-9.55
LF-774/0192	6.882	7.612	3.378	6.381	90.75	45.05	-9.61	-10.02
LF-774/0224	6.242	6.541	3.533	6.851	80.60	19.72	-9.68	-10.51
LF-774/0151	3.838	3.774	6.803	6.827	109.04	14.95	-9.50	-10.68
LF-774/0207	3.721	5.853	3.040	3.342	72.22	140.54	-9.74	-9.35

S12 Interface-confidence distributions for the initial design cohorts

The archived plots below describe the initial 1,000 designs from each generator, before sequence redesign. LigandAI is the platform through which LigandForge was accessed; these names refer to one cohort here [5]. AlphaProtein Novo supplied the separate cohort [15]. Each design was assessed as a Boltz-2 complex with human EGFR D3 and with mouse EGFR D3 [1]. Both analyses used ipSAE v4, the larger of the two chain-direction scores, with a 15 Å predicted-alignment-error cutoff [14]. The plots retain the archived densities and score values. Higher ipSAE indicates greater predicted interface confidence; it is not a measured affinity or evidence of pH selectivity.

S13 Completed rational batches and prediction coverage

Jobs 3880 and 3886 supplied 15 Novo single substitutions (one native model per species). Job 3871 supplied D46 WT, V8G and V8S (five native samples per species). Jobs 3963/3964 target TNF and are excluded from this EGFR analysis. All 60 EGFR native complexes have returned calibrated K_d and ΔG predictions from DeltaForge through LigandAI [5]; served K_d values are not recomputed from ΔG . The actual scorer and calibration versions, checkpoints, native sequences, source hashes and every sample-specific result are retained in the accompanying CSV and candidate packages. These are computational model assessments without experimental purification evidence. JustHISpKa 1.0.2 [8] assessed local EGFR chain B His37 and His100 on bound protein A+B, using AmberTools26 ff14SB and the retained NRC preparation. Four receptor disulfides were checked; native protein heavy atoms remained fixed without minimization. Native D46 glycan coordinates are retained in CIF files and omitted from protein-only scoring. Unforced and conditional predictions remain distinct; missing unforced estimates are not replaced by imposed acid-state values. Table S8 uses the confidence-selected sample independently per species. All five D46 samples, including the newly refolded WT control, remain in the package. The historical WT figure context is preserved; a new WT conformation does not count as a new sequence.

Table S8: New rational sequence predictions: served affinity and unforced receptor pKa, by species. K_d in nM; ΔG in kcal/mol; – means no unforced site estimate.

Candidate	K_d	Human			K_d	Mouse		
		ΔG	H37	H100		ΔG	H37	H100
Novo-0037_A56Q	78.60	-9.69	4.257	5.552	54.44	-9.91	4.931	6.465
Novo-0037_A56N	65.98	-9.80	4.647	6.501	102.53	-9.53	4.705	6.038
Novo-0037_A63S	83.83	-9.65	4.307	6.238	126.14	-9.41	4.524	6.804
Novo-0037_A63T	75.04	-9.72	4.574	6.926	101.67	-9.54	4.566	6.308
Novo-0037_A66S	94.19	-9.59	4.597	6.110	72.48	-9.74	4.760	6.360
Novo-0037_A66T	95.97	-9.57	4.282	6.678	99.17	-9.55	4.820	6.587
Novo-0037_A74S	158.28	-9.28	4.597	6.182	53.96	-9.92	4.690	6.225
Novo-0037_A74T	81.42	-9.67	4.430	6.106	101.56	-9.54	4.681	6.657
Novo-0470_A74S	4.11	-11.44	4.613	6.571	1.08	-12.23	4.390	9.209
Novo-0470_A74T	90.18	-9.61	4.339	6.652	1.08	-12.23	4.515	6.260
Novo-0470_L92S	188.79	-9.17	4.505	7.354	2.02	-11.86	4.465	6.782
Novo-0470_L92T	2.08	-11.84	4.241	5.772	0.77	-12.43	4.755	4.243
Novo-0470_L92M	4.30	-11.41	4.497	6.710	1.22	-12.16	4.638	6.453
Novo-0470_I27T	3.71	-11.50	4.500	5.572	3.34	-11.56	4.546	4.819
Novo-0470_I27M	1.76	-11.94	4.455	6.771	2.43	-11.75	4.483	3.945
D46-V8G	180.72	-9.20	6.816	7.579	148.25	-9.32	6.901	6.540
D46-V8S	150.07	-9.31	6.918	7.316	201.35	-9.14	7.048	6.959
D46-WT new control	141.59	-9.34	1.714	6.509	113.55	-9.47	3.028	5.336

The 17 rational additions, 100 LigandForge redesigns, 100 Novo redesigns and 100 Foundry originals extend the initial 77-candidate inventory to 394 exact unique sequences. All 394 have complete species-paired ipSAE, DeltaForge affinity and Potts predictions. The histidine-pKa maps use completed bound-complex JustHISpKa estimates, with 394 human unforced pairs and 392 mouse unforced pairs; the two unresolved mouse His100 cases retain separate labeled neighboring-acid-state outputs. Exact full sequences, native species/model hashes and raw calculation records link every result. Historical coverage tables and predictor-specific estimates remain preserved separately. No unavailable estimate was replaced with another predictor or an invented value.

S14 Species-specific confidence and predicted affinity

The two species-specific panels disambiguate the means in main Figure 3C. Both retain the same 394 binder sequences with complete paired affinity data, matching every main Figure 3 panel. Cohort identities and legend counts match the shared main-figure inventory. They show whether a mean combines similar predictions in both species or conceals a species difference. ipSAE measures confidence in the predicted interface [14]; DeltaForge through LigandAI supplies the predicted dissociation constant [5]. Lower K_d predicts stronger binding. Historical preparation and scorer settings limit comparisons between generators to descriptive assessments.

S15 Interface-confidence distributions after LigandForge sequence redesign

Supplementary Figure S5 complements the initial design distribution in Figure S3. It separates each starting binder's redesign library by receptor species, showing the full available distribution rather than only the ten variants selected per parent. The retained campaign contains 18,900 scored complexes from 9,450 matched pairs. Parent 774 is incomplete; its smaller sample is identified in the figure and preserved in the missing-pair ledger. No predictions were rerun or missing scores filled in for this figure.

S16 Expert redesign versus original: species-resolved changes

We compared all 37 expert DD1 redesigns and the two D46 DD2 V8 redesigns with their exact-sequence originals in each species (Figure S11). Each original was linked by its complete binder sequence and verified substitution set. ipSAE uses the retained larger chain-direction score with 15 Å PAE and distance cutoffs [14]. The differences are redesign minus original: $\Delta I_s = I_{r,s} - I_{p,s}$ and $\Delta K_{d,s} = K_{d,r,s} - K_{d,p,s}$ for species s . The plotted affinity change is $L_s = \log_2(K_{d,r,s}/K_{d,p,s})$. Thus positive ΔI_s indicates higher interface confidence, whereas negative $\Delta K_{d,s}$ or L_s indicates stronger predicted affinity. Served DeltaForge predictions are used directly [5]; K_d is not reconstructed from ΔG . The accompanying source tables also retain $\Delta\Delta G_s = \Delta G_{r,s} - \Delta G_{p,s}$, and species differences before and after redesign. Changes in species contrast are $\Delta I_M - \Delta I_H$, $\Delta K_{d,M} - \Delta K_{d,H}$ and $L_M - L_H$; these definitions avoid conflating a redesign effect with the original human/mouse difference.

D46 V8G/V8S are compared with the native WT refolded alongside them in job 3871, with every selected complex scored from its protein coordinates. All three use sample 0, the highest-confidence model independently in each species; all five samples per construct/species remain in the candidate package. The historical D46 sequence-scored control in the main figure remains a separate context. Earlier DD1 parents and redesigns have historical differences in templates, glycans and scorer settings; their deltas are descriptive, with no matched-preparation or causal claim. Predictions do not measure affinity changes, pH selectivity or uncertainty. No new prediction or structure inference was performed for this comparison. The inherited Foundry design lineage retains Foundry/AtomWorks/RF3 and RFD3 attribution [16, 6]; the Novo lineage uses AlphaProtein Novo [15]. The existing receptor-site coverage audit keeps JustHISpKa estimates distinct from PROPKA [8, 2].

Table S9: Species-resolved expert-parent deltas. ΔI : ipSAE redesign minus original; ΔK_d : redesign minus original in nM; L : $\log_2$ redesign/original K_d . H/M denote human/mouse. Rounded values are for display; exact values and parent baselines are retained in the source CSV.

Redesign	ΔI_H	ΔI_M	$\Delta K_{d,H}$	$\Delta K_{d,M}$	L_H	L_M
062 A73R	+0.087	+0.019	-48.28	-25.92	-0.37	-0.19
062 A76Q	+0.088	+0.091	-94.75	-121.48	-0.85	-1.27
062 I88Q	-0.009	-0.037	-59.39	+213.88	-0.47	+1.02
062 A73R+A76Q	+0.034	+0.081	-165.47	-86.46	-2.18	-0.78
062 A73R+I88Q	+0.144	+0.019	-70.55	-174.03	-0.58	-2.63
062 A76Q+I88Q	+0.125	+0.044	-104.96	-162.44	-0.98	-2.20
062 A73R+A76Q+I88Q	+0.157	+0.041	+432.87	+34.27	+1.60	+0.22
195 A7Q	+0.113	+0.109	+44.14	-31.42	+0.83	-0.82
195 A7R	+0.078	+0.116	+33.60	-10.03	+0.67	-0.21
195 F43Y	-0.034	+0.104	+267.14	+32.38	+2.51	+0.53
195 Q22Y	+0.106	-0.046	+23.30	+21.19	+0.50	+0.37
195 A7Q+F43Y	+0.063	+0.138	+106.47	+25.67	+1.52	+0.44
195 A7Q+Q22Y	+0.103	+0.147	+58.12	+39.31	+1.02	+0.63
195 A7R+F43Y	+0.037	+0.109	+23.44	-20.10	+0.50	-0.47
195 A7R+Q22Y	+0.031	+0.135	+127.18	-33.79	+1.70	-0.91
195 Q22Y+F43Y	-0.018	-0.039	+152.38	+70.67	+1.88	+0.98
195 A7Q+Q22Y+F43Y	-0.039	+0.134	+127.07	-37.11	+1.70	-1.03
195 A7R+Q22Y+F43Y	-0.221	+0.107	+186.11	+41.63	+2.10	+0.65
310 L18S	-0.055	+0.204	+308.93	-31.69	+2.39	-0.86
310 L18T	+0.147	-0.168	-0.97	-28.61	-0.02	-0.75
310 L18S+A32S	-0.283	+0.176	+33.44	-22.30	+0.54	-0.55
310 L18T+A32S	-0.131	-0.062	+49.97	-3.22	+0.75	-0.07
Novo-0037 A56N	+0.001	-0.007	+1.49	+1.30	+0.03	+0.02
Novo-0037 A56Q	-0.016	-0.013	+14.10	-46.78	+0.29	-0.89
Novo-0037 A63S	-0.007	+0.014	+19.34	+24.92	+0.38	+0.32
Novo-0037 A63T	-0.008	-0.015	+10.54	+0.45	+0.22	+0.01
Novo-0037 A66S	-0.005	+0.005	+29.69	-28.74	+0.55	-0.48
Novo-0037 A66T	+0.005	-0.015	+31.47	-2.06	+0.57	-0.03
Novo-0037 A74S	+0.002	-0.001	+93.78	-47.26	+1.30	-0.91
Novo-0037 A74T	-0.003	-0.015	+16.92	+0.34	+0.34	+0.00
Novo-0470 A74S	-0.002	-0.005	+0.02	-0.27	+0.01	-0.32
Novo-0470 A74T	-0.001	-0.011	+86.10	-0.27	+4.47	-0.32
Novo-0470 I27M	-0.004	-0.016	-2.32	+1.08	-1.21	+0.84
Novo-0470 I27T	+0.001	-0.020	-0.37	+1.98	-0.14	+1.30
Novo-0470 L92M	+0.007	-0.012	+0.22	-0.13	+0.08	-0.15
Novo-0470 L92S	-0.002	-0.013	+184.71	+0.66	+5.53	+0.58
Novo-0470 L92T	+0.000	-0.004	-2.00	-0.58	-0.97	-0.81
D46 V8G	-0.462	-0.681	+39.12	+34.70	+0.35	+0.38
D46 V8S	-0.382	-0.054	+8.47	+87.80	+0.08	+0.83

S17 Original Foundry top100 histidine characterization

For the original Foundry top 100, H37/H100 calculations were attempted in all 200 native human/mouse complexes with JustHISpKa 1.0.2, without externally imposed protonation forces [8]. Atom, topology, local-to-global numbering and source-hash audits were retained. Four canonical receptor disulfides plus any unambiguous native binder SG pairs below 2.5 Å were assigned; additional binder bonds are explicitly inferred. Hydrogens and terminal OXT were added with Amber ff14SB, without heavy-atom minimization. Twenty-nine contexts from 17 binders have inferred binder bonds. Six designs have different inferred assignments across species; these chemistry assumptions are preserved in the source-linked table. Local H37/H100 correspond to human source H346/H409 and mouse source H370/H433; the shared project-reference numbering is H346/H409. Global mean-ipSAE rank identifies each fixed selected design. Exact sequences and raw hashes are in the source-linked CSV/FASTA. –

denotes no unforced estimate; bold values meet the inclusive 6.5–7.4 interval before rounding. Species are not averaged.

Table S10: Foundry top100 EGFR H37/H100 JustHISpKa estimates.

Rank	Generation identity	Human H37	H100	Mouse H37	H100
1	Human_EGFR_Human_EGFR_295_model_1	6.802	6.251	6.820	3.901
2	Human_EGFR_Human_EGFR_537_model_5	4.913	2.199	5.159	4.997
3	Human_EGFR_Human_EGFR_650_model_5	4.861	5.801	7.261	6.370
4	design_DD1_Human_ASP_pair_8_model_1_16_model_1	5.325	7.793	5.925	8.353
5	Human_EGFR_Human_EGFR_75_model_5	4.056	2.350	7.063	5.734
6	Human_EGFR_Human_EGFR_315_model_7	3.728	1.089	5.168	5.589
7	Human_EGFR_Human_EGFR_1103_model_3	6.333	5.694	6.740	5.084
8	Human_EGFR_Human_EGFR_812_model_1	6.725	3.670	-0.946	2.019
9	Human_EGFR_Human_EGFR_821_model_6	2.992	4.588	5.889	6.445
10	design_DD1_Human_ASP_pair_17_model_1_5_model_3	6.597	5.234	4.484	5.860
11	Human_EGFR_Human_EGFR_203_model_2	3.839	5.880	7.470	5.167
12	Human_EGFR_Human_EGFR_1131_model_5	6.912	6.573	7.141	7.004
13	Human_EGFR_Human_EGFR_530_model_0	4.066	5.765	6.205	3.543
14	design_DD1_Human_ASP_pair_8_model_1_22_model_1	6.141	6.655	6.763	8.019
15	Human_EGFR_Human_EGFR_153_model_7	6.657	3.132	6.956	1.500
16	Human_EGFR_Human_EGFR_495_model_7	6.462	7.011	4.978	7.314
17	design_DD1_Human_ASP_pair_8_model_1_11_model_2	3.825	6.505	6.636	4.378
18	Human_EGFR_Human_EGFR_180_model_1	6.530	7.946	6.594	7.482
19	design_DD1_Human_ASP_pair_24_model_0_11_model_3	6.531	6.443	7.119	6.670
20	Human_EGFR_Human_EGFR_964_model_7	5.786	5.987	6.794	5.971
21	Human_EGFR_Human_EGFR_700_model_5	6.692	2.135	5.105	4.829
22	design_DD1_Human_ASP_pair_8_model_1_11_model_3	7.183	6.424	6.083	7.095
23	design_DD1_Human_ASP_pair_24_model_0_19_model_1	6.331	6.143	6.558	6.972
24	Human_EGFR_Human_EGFR_486_model_6	7.017	6.448	5.270	6.780
25	Human_EGFR_Human_EGFR_436_model_1	6.679	4.444	4.425	3.981
26	Human_EGFR_Human_EGFR_142_model_3	6.714	2.323	3.302	1.266
27	design_DD1_Human_ASP_pair_8_model_1_9_model_0	6.933	6.939	2.722	6.544
28	Human_EGFR_Human_EGFR_441_model_6	5.372	4.947	6.084	4.750
29	design_DD1_Human_ASP_pair_24_model_0_20_model_3	4.427	5.965	5.871	7.380
30	Human_EGFR_Human_EGFR_1103_model_5	6.658	6.401	6.656	6.932
31	Human_EGFR_Human_EGFR_582_model_1	4.661	7.570	4.459	6.801
32	Human_EGFR_Human_EGFR_486_model_7	4.341	6.297	4.198	7.673
33	Human_EGFR_Human_EGFR_180_model_3	3.184	4.282	2.277	6.788
34	Human_EGFR_Human_EGFR_1207_model_6	6.860	7.738	4.853	5.566
35	Human_EGFR_Human_EGFR_1083_model_0	6.526	6.050	5.166	6.775
36	design_DD1_Human_ASP_pair_11_model_1_11_model_2	7.076	3.545	4.714	6.394
37	Human_EGFR_Human_EGFR_258_model_5	1.783	6.907	3.667	6.936
38	Human_EGFR_Human_EGFR_1021_model_1	4.849	6.154	4.684	4.783
39	Human_EGFR_Human_EGFR_651_model_1	6.573	6.997	3.628	8.384
40	Human_EGFR_Human_EGFR_599_model_0	5.280	7.359	5.393	7.064
41	Human_EGFR_Human_EGFR_466_model_3	6.269	2.300	6.724	5.622
42	Human_EGFR_Human_EGFR_58_model_1	6.717	4.070	4.144	4.582
43	Human_EGFR_Human_EGFR_157_model_5	7.503	7.491	6.961	6.870
44	Human_EGFR_Human_EGFR_953_model_5	6.694	6.403	4.989	5.269
45	Human_EGFR_Human_EGFR_115_model_3	6.673	5.357	7.150	6.465
46	Human_EGFR_Human_EGFR_1156_model_3	4.631	6.894	4.791	6.609
47	Human_EGFR_Human_EGFR_1244_model_7	7.074	4.760	7.345	5.830
48	Human_EGFR_Human_EGFR_230_model_2	5.436	7.565	4.385	6.667
49	Human_EGFR_Human_EGFR_810_model_3	6.296	5.752	6.773	6.840
50	Human_EGFR_Human_EGFR_1185_model_1	6.959	7.779	5.099	6.450
51	Human_EGFR_Human_EGFR_420_model_1	4.519	6.123	6.801	4.937
52	Human_EGFR_Human_EGFR_1113_model_2	1.307	7.295	3.840	8.102
53	Human_EGFR_Human_EGFR_825_model_6	6.068	7.118	4.517	6.639
54	Human_EGFR_Human_EGFR_1004_model_2	6.794	6.472	6.329	4.628
55	design_DD1_Human_ASP_pair_8_model_1_12_model_1	6.400	6.588	3.252	4.559
56	Human_EGFR_Human_EGFR_35_model_1	3.455	5.849	6.715	5.460
57	Human_EGFR_Human_EGFR_874_model_7	3.793	4.491	5.141	6.859
58	Human_EGFR_Human_EGFR_882_model_1	6.912	7.989	7.159	5.524
59	Human_EGFR_Human_EGFR_603_model_6	4.248	2.835	4.880	4.509
60	Human_EGFR_Human_EGFR_50_model_7	6.975	6.965	7.169	5.731
61	Human_EGFR_Human_EGFR_1016_model_5	5.811	2.896	5.967	0.715
62	Human_EGFR_Human_EGFR_958_model_3	5.471	3.676	4.147	7.996
63	Human_EGFR_Human_EGFR_900_model_1	5.908	6.228	5.189	7.083
64	Human_EGFR_Human_EGFR_1051_model_7	4.295	7.490	4.045	7.153
65	Human_EGFR_Human_EGFR_689_model_6	3.673	4.708	4.195	2.875
66	Human_EGFR_Human_EGFR_1162_model_0	4.031	4.860	6.417	5.402
67	Human_EGFR_Human_EGFR_452_model_3	4.718	7.040	8.157	8.266
68	design_DD1_Human_ASP_pair_17_model_1_14_model_2	6.588	6.984	6.829	6.827
69	Human_EGFR_Human_EGFR_877_model_6	5.461	4.783	4.944	4.533
70	Human_EGFR_Human_EGFR_963_model_5	4.354	7.238	3.199	6.496
71	Human_EGFR_Human_EGFR_936_model_4	6.851	3.626	4.702	-0.296
72	Human_EGFR_Human_EGFR_403_model_6	3.121	6.617	3.554	5.275
73	Human_EGFR_Human_EGFR_428_model_1	7.181	6.666	6.645	7.231
74	Human_EGFR_Human_EGFR_895_model_0	6.373	7.267	4.272	7.683
75	Human_EGFR_Human_EGFR_1179_model_7	3.566	2.537	4.619	5.731

Rank	Generation identity	Human H37	H100	Mouse H37	H100
76	Human_EGFR_Human_EGFR_999_model_2	6.697	7.139	4.558	5.186
77	Human_EGFR_Human_EGFR_1160_model_4	6.742	5.257	3.656	7.098
78	Human_EGFR_Human_EGFR_676_model_5	4.984	7.432	4.843	6.989
79	Human_EGFR_Human_EGFR_597_model_1	6.742	7.417	4.667	5.609
80	Human_EGFR_Human_EGFR_1120_model_3	5.969	6.406	5.959	6.418
81	Human_EGFR_Human_EGFR_328_model_2	6.192	4.204	7.122	5.357
82	Human_EGFR_Human_EGFR_627_model_7	4.013	4.455	3.649	5.454
83	design_DD1_Human_ASP_pair_17_model_1_19_model_0	3.628	6.165	4.841	6.230
84	Human_EGFR_Human_EGFR_747_model_4	6.905	7.107	7.178	6.980
85	Human_EGFR_Human_EGFR_637_model_4	5.311	3.180	4.125	5.139
86	Human_EGFR_Human_EGFR_42_model_7	6.777	6.126	6.049	5.819
87	Human_EGFR_Human_EGFR_616_model_3	7.106	7.027	7.120	6.114
88	Human_EGFR_Human_EGFR_1012_model_6	6.598	4.294	4.888	6.126
89	design_DD1_Human_ASP_pair_17_model_1_10_model_2	5.296	6.869	7.409	6.532
90	Human_EGFR_Human_EGFR_160_model_2	4.731	5.010	4.495	6.880
91	Human_EGFR_Human_EGFR_410_model_1	7.344	7.365	5.250	7.727
92	Human_EGFR_Human_EGFR_745_model_7	7.214	7.114	3.866	7.336
93	Human_EGFR_Human_EGFR_104_model_0	7.448	5.519	7.529	6.219
94	Human_EGFR_Human_EGFR_755_model_0	6.808	3.952	7.413	4.797
95	design_DD1_Human_ASP_pair_17_model_1_5_model_0	6.703	5.018	4.704	4.982
96	Human_EGFR_Human_EGFR_606_model_7	4.662	6.503	5.752	8.144
97	Human_EGFR_Human_EGFR_527_model_2	5.545	6.500	6.913	4.457
98	Human_EGFR_Human_EGFR_96_model_6	6.997	7.901	4.534	7.571
99	Human_EGFR_Human_EGFR_126_model_4	7.200	7.001	7.110	6.678
100	Human_EGFR_Human_EGFR_925_model_5	6.805	7.128	3.796	5.815

S18 Original Foundry top100 affinity characterization

The fixed global mean-ipSAE selection is preserved. The following table gives the 100 returned mouse DeltaForge scores [5]. Kd is the served calibrated prediction in nM; no separate exponential conversion from deltaG was applied. A completed successor added 100 human native-complex predictions and paired them with these mouse results; the human and mouse values are included in main Figure 3 and Supplementary Figure S8. This historical mouse table is retained unchanged. Global rank and the source-linked CSV retain exact generation identities and sequences. Raw scorer version, checkpoint, PDB hashes and warnings remain linked from the CSV.

Table S11: Mouse EGFR DeltaForge predictions for the Foundry top100.

Rank	Generation identity	ΔG (kcal mol ⁻¹)	K_d (nM)
1	Human_EGFR_Human_EGFR_295_model_1	-10.275	29.4
2	Human_EGFR_Human_EGFR_537_model_5	-9.237	169
3	Human_EGFR_Human_EGFR_650_model_5	-9.517	106
4	design_DD1_Human_ASP_pair_8_model_1_16_model_1	-8.993	256
5	Human_EGFR_Human_EGFR_75_model_5	-10.284	29
6	Human_EGFR_Human_EGFR_315_model_7	-9.730	73.8
7	Human_EGFR_Human_EGFR_1103_model_3	-9.471	114
8	Human_EGFR_Human_EGFR_812_model_1	-8.952	274
9	Human_EGFR_Human_EGFR_821_model_6	-10.020	45.2
10	design_DD1_Human_ASP_pair_17_model_1_5_model_3	-10.346	26.1
11	Human_EGFR_Human_EGFR_203_model_2	-9.612	90.1
12	Human_EGFR_Human_EGFR_1131_model_5	-9.275	159
13	Human_EGFR_Human_EGFR_530_model_0	-10.602	16.9
14	design_DD1_Human_ASP_pair_8_model_1_22_model_1	-9.499	109
15	Human_EGFR_Human_EGFR_153_model_7	-11.617	3.05
16	Human_EGFR_Human_EGFR_495_model_7	-11.209	6.08
17	design_DD1_Human_ASP_pair_8_model_1_11_model_2	-9.348	141
18	Human_EGFR_Human_EGFR_180_model_1	-9.193	182
19	design_DD1_Human_ASP_pair_24_model_0_11_model_3	-10.121	38.1
20	Human_EGFR_Human_EGFR_964_model_7	-9.434	122
21	Human_EGFR_Human_EGFR_700_model_5	-11.162	6.58
22	design_DD1_Human_ASP_pair_8_model_1_11_model_3	-9.522	105
23	design_DD1_Human_ASP_pair_24_model_0_19_model_1	-10.133	37.4
24	Human_EGFR_Human_EGFR_486_model_6	-9.610	90.3
25	Human_EGFR_Human_EGFR_436_model_1	-9.495	110
26	Human_EGFR_Human_EGFR_142_model_3	-8.533	556
27	design_DD1_Human_ASP_pair_8_model_1_9_model_0	-10.034	44.2
28	Human_EGFR_Human_EGFR_441_model_6	-9.874	57.9
29	design_DD1_Human_ASP_pair_24_model_0_20_model_3	-9.606	91
30	Human_EGFR_Human_EGFR_1103_model_5	-9.504	108
31	Human_EGFR_Human_EGFR_582_model_1	-9.461	116
32	Human_EGFR_Human_EGFR_486_model_7	-10.037	44
33	Human_EGFR_Human_EGFR_180_model_3	-9.821	63.2
34	Human_EGFR_Human_EGFR_1207_model_6	-9.570	96.7
35	Human_EGFR_Human_EGFR_1083_model_0	-9.480	113
36	design_DD1_Human_ASP_pair_11_model_1_11_model_2	-10.088	40.3

Rank	Generation identity	ΔG	K_d
37	Human_EGFR_Human_EGFR_258_model_5	-9.892	56.1
38	Human_EGFR_Human_EGFR_1021_model_1	-8.690	427
39	Human_EGFR_Human_EGFR_651_model_1	-9.328	145
40	Human_EGFR_Human_EGFR_599_model_0	-9.726	74.3
41	Human_EGFR_Human_EGFR_466_model_3	-10.480	20.8
42	Human_EGFR_Human_EGFR_58_model_1	-8.481	607
43	Human_EGFR_Human_EGFR_157_model_5	-12.029	1.52
44	Human_EGFR_Human_EGFR_953_model_5	-9.375	134
45	Human_EGFR_Human_EGFR_115_model_3	-10.064	42
46	Human_EGFR_Human_EGFR_1156_model_3	-10.370	25
47	Human_EGFR_Human_EGFR_1244_model_7	-9.464	115
48	Human_EGFR_Human_EGFR_230_model_2	-9.544	101
49	Human_EGFR_Human_EGFR_810_model_3	-9.879	57.4
50	Human_EGFR_Human_EGFR_1185_model_1	-9.491	110
51	Human_EGFR_Human_EGFR_420_model_1	-9.541	101
52	Human_EGFR_Human_EGFR_1113_model_2	-9.543	101
53	Human_EGFR_Human_EGFR_825_model_6	-9.570	96.7
54	Human_EGFR_Human_EGFR_1004_model_2	-9.944	51.4
55	design_DD1_Human_ASP_pair_8_model_1_12_model_1	-10.106	39.1
56	Human_EGFR_Human_EGFR_35_model_1	-9.846	60.7
57	Human_EGFR_Human_EGFR_874_model_7	-9.930	52.6
58	Human_EGFR_Human_EGFR_882_model_1	-9.647	84.8
59	Human_EGFR_Human_EGFR_603_model_6	-9.246	167
60	Human_EGFR_Human_EGFR_50_model_7	-9.167	191
61	Human_EGFR_Human_EGFR_1016_model_5	-9.613	89.8
62	Human_EGFR_Human_EGFR_958_model_3	-10.848	11.2
63	Human_EGFR_Human_EGFR_900_model_1	-10.079	40.9
64	Human_EGFR_Human_EGFR_1051_model_7	-9.366	136
65	Human_EGFR_Human_EGFR_689_model_6	-9.942	51.6
66	Human_EGFR_Human_EGFR_1162_model_0	-9.364	137
67	Human_EGFR_Human_EGFR_452_model_3	-8.920	290
68	design_DD1_Human_ASP_pair_17_model_1_14_model_2	-9.645	85.2
69	Human_EGFR_Human_EGFR_877_model_6	-9.328	145
70	Human_EGFR_Human_EGFR_963_model_5	-9.230	172
71	Human_EGFR_Human_EGFR_936_model_4	-10.260	30.2
72	Human_EGFR_Human_EGFR_403_model_6	-9.299	153
73	Human_EGFR_Human_EGFR_428_model_1	-10.203	33.2
74	Human_EGFR_Human_EGFR_895_model_0	-8.683	432
75	Human_EGFR_Human_EGFR_1179_model_7	-9.493	110
76	Human_EGFR_Human_EGFR_999_model_2	-10.182	34.4
77	Human_EGFR_Human_EGFR_1160_model_4	-9.566	97.3
78	Human_EGFR_Human_EGFR_676_model_5	-9.327	146
79	Human_EGFR_Human_EGFR_597_model_1	-8.996	255
80	Human_EGFR_Human_EGFR_1120_model_3	-10.664	15.2
81	Human_EGFR_Human_EGFR_328_model_2	-10.248	30.8
82	Human_EGFR_Human_EGFR_627_model_7	-9.637	86.4
83	design_DD1_Human_ASP_pair_17_model_1_19_model_0	-9.557	98.8
84	Human_EGFR_Human_EGFR_747_model_4	-9.850	60.2
85	Human_EGFR_Human_EGFR_637_model_4	-9.169	190
86	Human_EGFR_Human_EGFR_42_model_7	-9.103	213
87	Human_EGFR_Human_EGFR_616_model_3	-9.088	218
88	Human_EGFR_Human_EGFR_1012_model_6	-9.705	76.9
89	design_DD1_Human_ASP_pair_17_model_1_10_model_2	-10.131	37.5
90	Human_EGFR_Human_EGFR_160_model_2	-9.919	53.6
91	Human_EGFR_Human_EGFR_410_model_1	-9.388	131
92	Human_EGFR_Human_EGFR_745_model_7	-10.020	45.2
93	Human_EGFR_Human_EGFR_104_model_0	-9.704	77.1
94	Human_EGFR_Human_EGFR_755_model_0	-9.033	239
95	design_DD1_Human_ASP_pair_17_model_1_5_model_0	-10.215	32.5
96	Human_EGFR_Human_EGFR_606_model_7	-9.454	117
97	Human_EGFR_Human_EGFR_527_model_2	-9.791	66.5
98	Human_EGFR_Human_EGFR_96_model_6	-9.441	120
99	Human_EGFR_Human_EGFR_126_model_4	-10.033	44.2
100	Human_EGFR_Human_EGFR_925_model_5	-12.380	0.842

S19 Ten complete AlphaProtein Novo redesign cohorts

The final verified job-3835 collection contains all 10,000 exact human/mouse pairs: 1,000 pairs for each of the ten original Novo parents. The ten highest descending arithmetic mean human/mouse ipSAE values per parent were retained, using saved design identity to resolve ties, yielding 100 distinct sequences and 200 complexes. The original 70 selections are preserved; parents 0417, 0754 and 0026 supply 30 additional selections after each full library passed native-file, sequence and pairing validation. Every selection links to its accepted-generation record and native model hashes. Affinity and ionization predictions did not influence this selection. Standalone Proton-PottsMPNN v6 [12, 13] generated redesigns from AlphaProtein Novo [15] starting binders. RosettaCommons Foundry [6] is related-framework attribution rather than the recorded redesign execution framework.

JustHISpKa 1.0.2 [8] assessed the exact native-fragment receptor sites B:37/B:100 in each species-specific bound protein complex (project full-EGFR mapping H346/H409). AmberTools ff14SB preparation preserved protein heavy coordinates without minimization, retained the four EGFR disulfides and exact Amber-detected unambiguous intra-binder bonds, and audited histidine mappings and topology. All 200 source complexes lack glycans. All 400 requested unforced receptor-site estimates completed. Original 70-pair predictions were reused unchanged; the additional 30 pairs used the same validated preparation and scorer.

All 200 native protein PDB requests completed through DeltaForge via LigandAI [5]. Served calibrated K_d values in nM are retained directly; they were not derived from ΔG . The model accepted no pH input. Scorer versions, calibration heads, input hashes and full response records remain linked to each species observation. These predictions do not establish measured affinity or pH selectivity.

Table S12: Top ten Novo-derived variants per complete parent. Human/mouse predictions are paired in each cell. I : ipSAE; K_d : nM; ΔG : kcal/mol; H37/H100: unforced EGFR D3 pKa. Variant names preserve parent/variant numbering.

Parent/variant	Rank	Mean I	H/M I	H/M K_d	H/M ΔG	H/M H37	H/M H100
0026/0685	1	0.941510	0.946364	677.9	-8.416	6.877	5.772
			0.936656	3.015	-11.624	7.058	6.009
0026/0445	2	0.940415	0.945304	90.42	-9.609	6.643	5.667
			0.935525	4.409	-11.399	6.731	5.564
0026/0593	3	0.939845	0.928128	3.509	-11.534	6.916	5.239
			0.951562	1.697	-11.965	7.157	5.450
0026/0383	4	0.939139	0.939106	115.2	-9.466	7.123	6.035
			0.939172	3.428	-11.548	6.968	6.131
0026/0576	5	0.939041	0.932948	182.9	-9.192	7.012	5.777
			0.945134	2.098	-11.839	7.059	5.885
0026/0456	6	0.938507	0.944654	28.96	-10.284	6.797	5.433
			0.932360	56.27	-9.890	7.013	5.451
0026/0646	7	0.938470	0.941488	127.8	-9.405	6.881	5.521
			0.935453	3.099	-11.608	7.380	5.594
0026/0033	8	0.938356	0.944454	53.11	-9.925	6.731	5.784
			0.932258	305	-8.889	6.740	5.918
0026/0287	9	0.938148	0.946825	98.81	-9.557	6.923	5.725
			0.929471	69.77	-9.763	6.887	5.826
0026/0732	10	0.937982	0.949404	4.517	-11.385	6.526	5.598
			0.926560	6.956	-11.129	6.957	5.560
0037/0957	1	0.949322	0.953652	72.14	-9.743	4.434	6.778
			0.944992	46.67	-10.001	4.530	6.581
0037/0225	2	0.949027	0.959277	46.67	-10.001	4.508	6.718
			0.938777	72.45	-9.741	4.305	4.471
0037/0162	3	0.945615	0.948416	285.2	-8.929	4.641	6.293
			0.942814	85.69	-9.641	4.947	6.005
0037/0111	4	0.944919	0.946636	40.24	-10.089	4.580	5.572
			0.943202	39.2	-10.105	4.701	2.497
0037/0688	5	0.944751	0.948607	57.27	-9.880	5.138	5.568
			0.940896	52.35	-9.933	4.595	4.364
0037/0725	6	0.944482	0.944476	37.5	-10.131	4.184	5.843
			0.944488	36.9	-10.140	4.625	5.372
0037/0120	7	0.944392	0.948014	63.12	-9.822	4.682	4.738
			0.940769	89.61	-9.615	4.616	4.541
0037/0483	8	0.944242	0.943752	53.84	-9.917	4.860	5.775
			0.944732	74.53	-9.724	4.833	4.853

Parent/variant	Rank	Mean I	H/M I	H/M K_d	H/M ΔG	H/M H37	H/M H100
0037/0651	9	0.944141	0.946166	82.79	-9.662	4.716	6.011
			0.942116	52.67	-9.930	4.636	6.154
			0.931677	59.38	-9.859	4.510	6.229
0037/0655	10	0.943405	0.955133	26.81	-10.330	4.384	5.930
			0.949715	19.7	-10.512	6.677	5.342
0293/0596	1	0.946473	0.943232	129.4	-9.397	6.639	6.482
			0.954592	41.97	-10.064	6.919	7.404
0293/0049	2	0.946050	0.937508	46.47	-10.004	4.524	6.828
			0.951141	29.81	-10.267	2.359	8.134
0293/0550	3	0.946032	0.940922	47.83	-9.987	6.530	7.496
			0.950899	20.58	-10.486	6.504	6.813
0293/0978	4	0.945655	0.940410	51.17	-9.947	3.708	7.285
			0.944117	45.76	-10.013	6.506	6.144
0293/0146	5	0.944376	0.944635	89.17	-9.618	3.559	6.555
			0.942932	82.25	-9.666	5.552	5.972
0293/0488	6	0.944357	0.945782	28.78	-10.288	7.010	6.078
			0.946309	45.85	-10.012	6.877	4.681
0293/0710	7	0.944265	0.942222	26.22	-10.343	2.895	5.523
			0.946517	65.82	-9.798	6.413	6.298
0293/0876	8	0.943832	0.941147	27.35	-10.318	3.751	7.816
			0.947192	36.95	-10.140	4.116	6.907
0293/0609	9	0.943822	0.940452	131.1	-9.390	5.062	7.155
			0.947157	44.66	-10.027	4.250	7.101
0293/0951	10	0.943664	0.940171	69.61	-9.764	7.089	6.385
			0.958963	19.33	-10.523	3.936	7.081
			0.945332	91.45	-9.603	3.960	6.330
0394/0408	1	0.952147	0.950599	531.8	-8.560	2.734	8.573
			0.950402	20.82	-10.480	3.455	6.367
0394/0029	2	0.950500	0.961002	24.43	-10.385	2.570	5.600
			0.939364	153.3	-9.297	2.247	6.875
0394/0471	3	0.950183	0.955647	112.9	-9.478	2.983	7.287
			0.942651	10.54	-10.883	3.327	6.311
0394/0132	4	0.949149	0.950211	1.915	-11.893	3.347	5.680
			0.947949	48.57	-9.978	4.048	6.877
0394/0783	5	0.949080	0.957679	30.94	-10.245	3.750	6.869
			0.940247	19.69	-10.513	3.820	6.956
0394/0388	6	0.948963	0.954754	77.58	-9.700	3.958	7.013
			0.942395	1.4	-12.079	-0.260	5.539
0394/0871	7	0.948574	0.949882	44	-10.036	1.509	6.589
			0.947164	1032	-8.167	4.208	7.116
0394/0184	8	0.948523	0.946888	222.2	-9.077	1.654	7.289
			0.949923	46.46	-10.004	3.184	4.925
0394/0466	9	0.948406	0.954345	81.01	-9.674	3.342	6.390
			0.942036	109.7	-9.495	3.628	6.637
0394/0064	10	0.948190	0.949115	57.5	-9.878	3.490	5.206
			0.945437	69.4	-9.766	3.800	4.792
0417/0868	1	0.947276	0.951165	3.238	-11.582	4.108	5.912
			0.935598	64.51	-9.809	4.248	5.109
0417/0764	2	0.943382	0.945508	44.33	-10.032	3.934	5.828
			0.939616	40.92	-10.079	4.232	5.950
0417/0328	3	0.942562	0.954137	41.67	-10.068	3.539	5.212
			0.928831	35.4	-10.165	4.011	5.210
0417/0260	4	0.941484	0.954977	78.09	-9.696	5.207	6.240
			0.926302	45.33	-10.018	4.261	5.655
0417/0120	5	0.940639	0.943898	107.2	-9.509	4.659	6.220
			0.937287	262.1	-8.979	4.714	6.194
0417/0121	6	0.940592	0.950358	49.79	-9.963	3.482	5.366
			0.929544	62.62	-9.827	3.723	5.251
0417/0596	7	0.939951	0.942122	35.28	-10.167	4.060	5.459
			0.933994	50.55	-9.954	3.742	5.677
0417/0739	8	0.938058	0.944406	220.3	-9.082	4.025	5.794
			0.930554	38.01	-10.123	3.656	5.957
0417/0736	9	0.937480	0.937531	430.6	-8.685	3.887	5.796
			0.937104	78.9	-9.690	5.301	5.394
0417/0648	10	0.937318	0.952213	5.499	-11.268	3.419	6.478
			0.946847	117.3	-9.455	4.738	6.117
0470/0082	1	0.949530	0.950766	233.3	-9.048	4.133	7.422
			0.941790	111.7	-9.484	4.445	6.604
0470/0318	2	0.946278	0.950558	180	-9.201	4.482	5.066
			0.940113	1.262	-12.140	4.488	5.422
0470/0090	3	0.945335	0.948403	1.657	-11.979	4.277	6.891
			0.941503	59.8	-9.854	4.556	7.563
0470/0673	4	0.944953	0.946389	1.305	-12.120	4.734	5.495
			0.943141	1.818	-11.924	4.659	6.304

Parent/variant	Rank	Mean I	H/M I	H/M K_d	H/M ΔG	H/M H37	H/M H100
0470/0739	6	0.944192	0.945685	4.924	-11.334	3.339	4.522
			0.942700	891.3	-8.254	4.532	4.759
			0.945275	483.8	-8.616	4.398	6.478
0470/0748	7	0.943980	0.942685	1.154	-12.193	4.829	6.226
			0.941005	1.442	-12.061	4.195	6.673
0470/0039	8	0.942867	0.944729	443.1	-8.668	4.377	7.257
			0.945013	1.765	-11.942	4.432	4.192
0470/0804	9	0.942133	0.939252	0.9078	-12.336	4.433	8.135
			0.945831	194.6	-9.155	4.554	4.587
0470/0425	10	0.941943	0.938054	53.22	-9.923	4.437	2.641
			0.957561	1.586	-12.005	3.838	6.399
0599/0221	1	0.952431	0.947301	1.017	-12.268	4.224	6.170
			0.957019	0.8546	-12.371	3.981	6.498
0599/0376	2	0.952203	0.947387	1.435	-12.064	3.929	5.856
			0.957503	1.48	-12.046	4.851	5.671
0599/0437	3	0.951075	0.944648	0.7979	-12.412	4.512	6.119
			0.952474	1.89	-11.901	4.093	5.662
0599/0226	4	0.950613	0.948751	1.951	-11.882	4.456	5.333
			0.957896	1.908	-11.895	4.205	5.577
0599/0052	5	0.949855	0.941813	1.131	-12.205	4.146	6.214
			0.950370	0.9489	-12.309	3.995	5.447
0599/0931	6	0.948953	0.947535	1.755	-11.945	4.157	6.159
			0.949312	3.62	-11.516	4.120	5.307
0599/0540	7	0.948805	0.948298	1.922	-11.891	4.384	5.737
			0.957308	1.133	-12.204	4.268	4.467
0599/0656	8	0.948765	0.940222	2.397	-11.760	4.485	5.421
			0.952344	1.129	-12.206	3.909	6.411
0599/0101	9	0.948765	0.945185	0.8931	-12.345	4.540	5.060
			0.949127	1.021	-12.266	4.019	5.202
0599/0714	10	0.948312	0.947497	0.6969	-12.492	4.032	5.821
			0.955075	95.97	-9.574	7.147	4.629
0754/0905	1	0.951554	0.948033	81.84	-9.668	7.287	4.444
			0.950199	139.1	-9.354	6.893	4.391
0754/0350	2	0.948824	0.947448	49.45	-9.967	6.695	6.569
			0.940139	1.128	-12.207	7.312	6.799
0754/0129	3	0.945831	0.951522	2.02	-11.862	7.340	8.425
			0.951332	219.8	-9.083	7.384	6.083
0754/0133	4	0.945199	0.939067	19.79	-10.510	7.671	7.029
			0.958161	165.6	-9.251	6.717	7.887
0754/0769	5	0.944881	0.931601	1.006	-12.274	7.053	7.655
			0.953736	147.2	-9.321	7.308	5.165
0754/0874	6	0.944485	0.935234	102.8	-9.534	7.029	4.882
			0.950851	2.958	-11.636	7.045	6.106
0754/0607	7	0.943414	0.935976	2.052	-11.852	7.047	6.569
			0.949644	75.31	-9.718	6.901	7.336
0754/0223	8	0.943406	0.937167	1.001	-12.278	6.984	6.876
			0.942825	50.04	-9.960	6.781	7.841
0754/0313	9	0.943360	0.943894	2.911	-11.645	6.577	7.860
			0.946528	107.5	-9.507	7.275	7.799
0754/0119	10	0.942959	0.939390	1.718	-11.957	7.444	6.033
			0.950609	128.2	-9.403	4.964	4.503
0860/0598	1	0.945240	0.939871	143.4	-9.336	4.742	3.144
			0.939203	1.302	-12.122	5.707	5.474
0860/0576	2	0.940316	0.941428	2.163	-11.821	5.794	6.042
			0.936854	107.5	-9.507	4.884	3.950
0860/0401	3	0.936379	0.935905	2.744	-11.680	4.670	4.683
			0.937535	286.8	-8.925	4.957	4.562
0860/0538	4	0.934491	0.931447	135.6	-9.369	4.845	4.985
			0.935217	581.9	-8.506	5.780	8.199
0860/0380	5	0.932744	0.930271	719.8	-8.380	5.462	8.432
			0.953579	247.7	-9.012	3.021	7.132
0860/0109	6	0.932719	0.911858	71.19	-9.751	4.909	6.571
			0.932201	1.124	-12.209	4.258	4.781
0860/0663	7	0.932338	0.932476	37.66	-10.128	5.402	5.209
			0.936814	330.9	-8.841	4.418	5.099
0860/0222	8	0.930611	0.924408	21.25	-10.467	5.142	4.258
			0.932398	183.5	-9.190	5.099	4.127
0860/0330	9	0.930594	0.928791	3.114	-11.605	4.595	4.311
			0.933392	704.2	-8.393	5.703	4.719
0860/0210	10	0.930578	0.927763	97.09	-9.567	7.977	5.753
			0.957162	9.242	-10.961	3.604	7.317
0960/0988	1	0.953662	0.950162	87.99	-9.626	4.946	6.945
			0.955768	2.689	-11.692	3.132	7.089
0960/0697	2	0.952557	0.949345	35.18	-10.169	3.466	6.050

Parent/variant	Rank	Mean I	H/M I	H/M K_d	H/M ΔG	H/M H37	H/M H100
0960/0597	3	0.951391	0.950991	1.672	-11.974	2.911	7.161
			0.951791	1.177	-12.182	4.115	7.190
			0.948861	1.182	-12.179	3.598	7.000
0960/0425	4	0.950359	0.951857	0.6215	-12.560	4.856	7.050
			0.959591	0.8816	-12.353	4.379	6.493
0960/0502	5	0.949356	0.939122	0.7356	-12.460	3.756	6.541
			0.950127	1.05	-12.249	4.205	6.965
0960/0750	6	0.948664	0.947201	0.6115	-12.570	4.011	6.933
			0.950589	1.009	-12.273	3.990	6.905
0960/0795	7	0.948554	0.946519	13.55	-10.734	4.079	7.124
			0.957296	1.575	-12.009	2.921	7.188
0960/0273	8	0.948079	0.938861	0.9404	-12.315	3.411	7.244
			0.949750	0.9249	-12.324	3.522	7.083
0960/0119	9	0.947799	0.945848	1.848	-11.914	3.614	7.314
			0.956971	1.319	-12.114	2.617	7.362
0960/0386	10	0.947575	0.938179	1.112	-12.215	3.320	7.088

Human values appear above mouse values in each paired cell. Full sequences, immutable parent/selection links, ranks, source hashes and unrounded results are retained in the selected-100 paired score table; species/site tables retain all 200 complexes and 400 histidine estimates.

S20 Initial and redesigned AlphaProtein Novo interface-confidence distributions

Figure S6 brings together the initial 1,000-design Novo cohort and all ten complete 1,000-pair redesign libraries before selection. The initial cohort is the same population shown in Figure S4, replotted with a common fixed smoothing method. Redesign panels follow the paired human/mouse presentation used for LigandForge in Figure S5. Both species are matched by exact design identity; the paired mean is computed per design before ranking. Every parent contains 1,000 pairs, totaling 10,000 redesigns and 20,000 complexes. All 100 top-ten selections contribute to every main Figure 3 panel, with source-linked completed affinity, receptor-pKa and independent-site Potts predictions. Initial and redesigned campaigns differ in preparation settings, so the distributions support a descriptive comparison.

S21 Panel-specific five-panel assessment and full candidate maps

Main Figure 3 compares the same 394 exact binder sequences in every panel: 392 with complete unforced data and two with unresolved mouse His100 displayed under both retained neighboring-acid assumptions. All 394 have finite paired native ipSAE and Potts scores, positive paired DeltaForge K_d , and complete human receptor pKas. The conditional cases retain blank unforced mouse His100 fields. All 100 Novo redesigns now have complete follow-up results and are included throughout. Candidate matching uses exact full sequences; no missing observation or unforced pKa was imputed.

Table S13: Eligibility for all five main Figure 3 panels. The two conditional cases are included with explicit acid-state labels. Each count refers to unique exact binder sequences.

Cohort	Full inventory	All main panels
Novo originals	10	10
LigandForge originals	10	10
Foundry originals	100	100
DD1 redesigns	30	30
Expert redesigns	37	37
DD1 parents	3	3
D46 constructs	4	4
LigandForge redesigns	100	100
Novo redesigns	100	100
Total	394	394

The expanded Potts calculation scored all 788 native human/mouse complexes on CPU, without new folding or sequence generation. The 230 redesigned sequences used their recorded design centers. The other 164 used up to two binder H/D/E sites selected by shortest side-chain-heavy-atom distance to receptor heavy atoms in the human model; mouse retained those exact sequence positions. No distance cutoff was imposed. The farthest selected center was 28.745 Å from the target, so selected centers must not all be interpreted as direct interface contacts. Coordinates and all non-varied protonation labels were fixed. Protein-only coordinates omitted non-protein hetero atoms. HBPLUS-derived features [11] supported the retained protonation-labeling and standalone Proton-PottsMPNN context [12, 13, 6]; PROPKA estimated the varied binder-center pKas. The arithmetic independent-site Henderson–Hasselbalch weights used pH 6.5 and 7.4 and common pKa shifts of -1 , 0 and $+1$. Two archived control complexes reproduced their prior pKas and microstate scores exactly. This expanded score does not directly vary receptor His37/His100, model DD2 unfolding, or calculate a binding free energy.

Supplementary Data 3A–3E provide separate files for the five main panels and retain every panel metric, exact sequence, stable candidate identity, campaign origin and native model identity. Files A–D have 394 rows. File E has 396 rows for the same 394 sequences: the two conditional cases each have separate acid-deprotonated and acid-protonated output identifiers and plotted values. The unforced mouse His100 field remains blank for both. Conditional values, force options and raw source hashes are retained in every file. These assumptions are neither averaged nor assigned to assay endpoints. Window flags distinguish unforced and conditional values; an unforced-complete eligibility field remains false for these two candidates.

S21.1 Conditional mouse His100 predictions retained in the plots

The mouse DD1-062 A76Q+I88Q calculation varied only binder Asp55: FORCE_ASP=55 gave 6.771 and FORCE_ASH=55 gave 4.716. The mouse D46 fusion calculation fixed Glu16 and Asp60 together: FORCE_GLU=16, FORCE_ASP=60 gave 8.173; FORCE_GLU=16, FORCE_ASH=60 gave 2.338. Local EGFR B:His100 maps to global prepared sites 189 and 191, respectively. These outputs use JustHISpKa 1.0.2 [8], the retained AmberTools26/ff14SB preparation and unchanged protein coordinates. The unforced calculations reported multiple titratable groups and did not return a single pKa. Previously completed outputs and raw hashes were verified for this update; no new inference was run. The two acid assumptions are not exhaustive protonation ensembles (D46 has additional mixed possibilities) and are not assigned to pH 6.5/7.4. Labels $-H$ and $+H$ refer only to the imposed nearby-acid states. Panel E and the full mouse map each have 396 plotted outputs for 394 sequences.

The source-linked CSV inventory retains the exact sequences, centers, distances, species-specific outcomes, pKa method and scope, and the reason for each main-panel exclusion. All plotted receptor pKas use the source-linked bound-complex JustHISpKa successor; older PROPKA and receptor-only estimates remain in preserved historical tables and snapshots. Neither the complete-data filter nor an in-window receptor pKa validates pH-sensitive binding.

Complete pH accounting. All 394 selected sequences have complete Potts results for human and mouse EGFR (788 complexes). Both the main and full inventories contain 104 candidates acid-favored in both species at nominal pKas and 96 under every common shift. Table S14 gives cohort counts. Supplementary Data 4 preserve the 394-row candidate ledger, 788-row species ledger with both endpoint expectations at every shift, all protonation-pattern scores and source hashes. Endpoint expectations are independently reproduced from retained patterns and binder-center pKas. These predictions do not cover every generated library member and do not calculate binding free energy.

Table S14: Complete Potts accounting for the selected candidate inventory shared by all main Figure 3 panels. Acid-favored means strictly positive $S(7.4) - S(6.5)$ in both species. Shift-stable requires this direction at every common binder-site pKa shift.

Cohort	N	Acid	Stable
Novo originals	10	7	6
LigandForge originals	10	7	6
DD1 redesigns	30	5	3
Expert redesigns	37	17	17
DD1 parents	3	3	3
D46 constructs	4	2	2
Foundry originals	100	43	39
LigandForge redesigns	100	6	6
Novo redesigns	100	14	14
Total	394	104	96

S22 Supplementary figures

Figures are presented in the order in which they are first cited in the main manuscript, including its figure captions. Each figure has a linked entry in the table of contents. Figures S8 and S10 include the completed affinity results and explicitly conditional His100 outputs described in section S21.

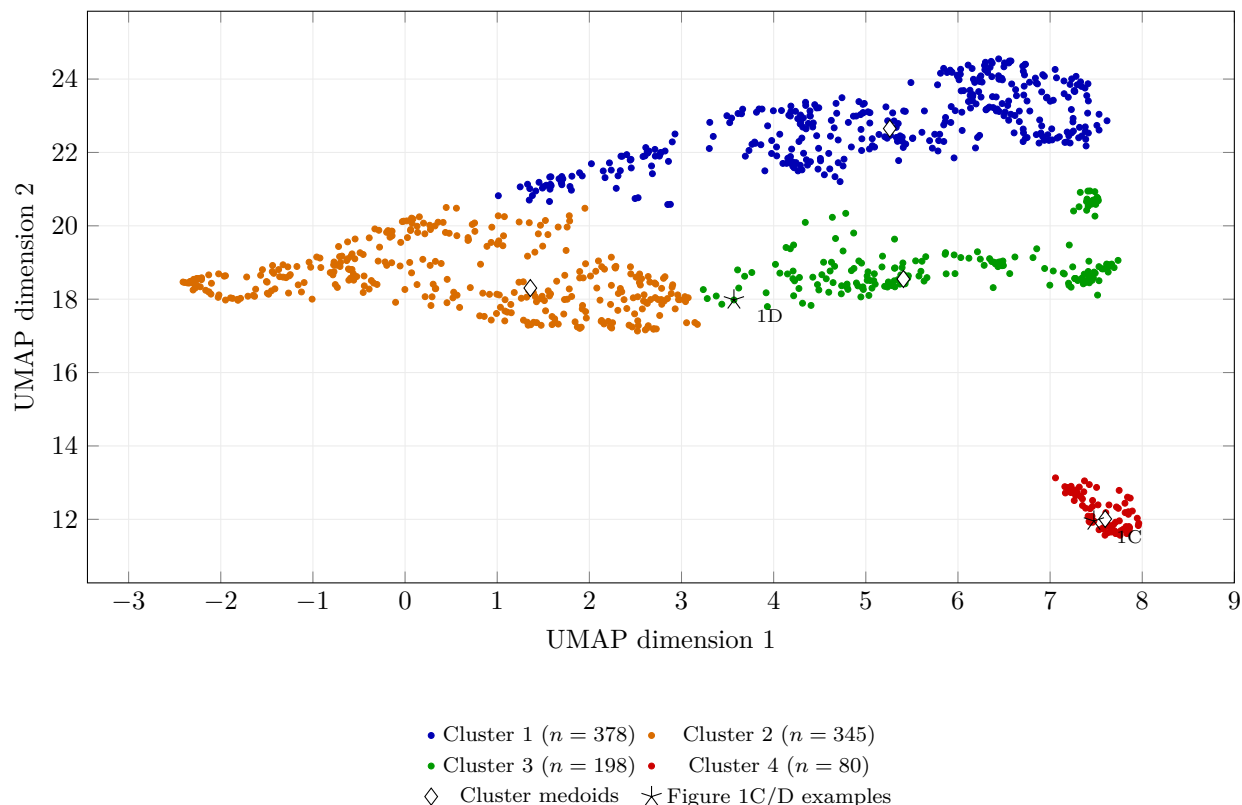
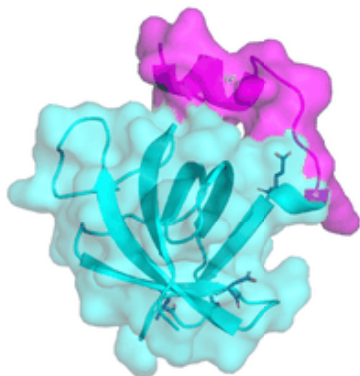


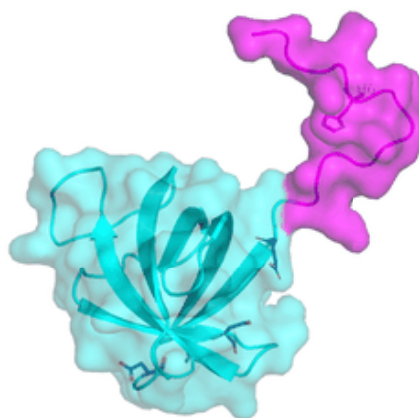
Figure S1: Recovered UMAP projection and four geometric clusters for the D46 AFSample3 workflow. For dimensional reduction, each fusion model was represented by a feature vector containing the upper-triangular, off-diagonal entries of its $C\alpha$ - $C\alpha$ distance matrix, using all $C\alpha$ atoms in the DD1-DD2 fusion. These features record every unique residue-pair distance within and between the two designed domains. The 1,001 plotted entries comprise 1,000 predictions plus a run-level entry. Colors and cluster membership follow the saved analysis. Open diamonds identify the four actual full-cluster $C\alpha$ RMSD medoids; stars labeled 1C/1D identify the selected main Figure 1 examples. The latter are not the medoids. The medoid models are shown in Figure S2, with identifiers and counts in Table S6. No cluster is labeled as a masked or exposed state, and cluster counts are not equilibrium populations.

A · Cluster 1 medoid

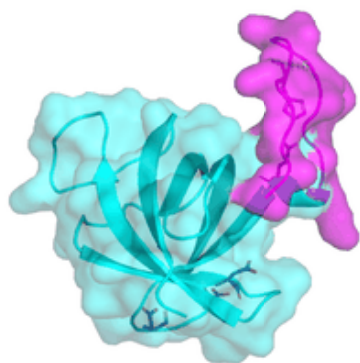
Seed 166 / sample 3 · 378 entries

**B · Cluster 2 medoid**

Seed 121 / sample 3 · 345 entries

**C · Cluster 3 medoid**

Seed 83 / sample 4 · 198 entries

**D · Cluster 4 medoid**

Seed 176 / sample 4 · 80 entries

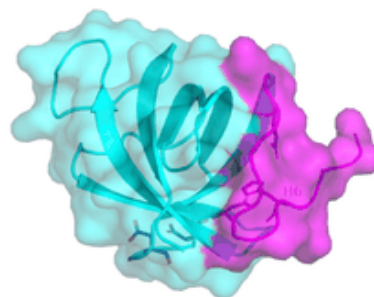


Figure S2: The four archived D46 AFSample3 medoid structures. (A) Cluster 1, seed 166/sample 3; (B) cluster 2, seed 121/sample 3; (C) cluster 3, seed 83/sample 4; (D) cluster 4, seed 176/sample 4. DD1 is cyan and DD2 magenta, with semi-transparent surfaces over cartoons. Highlighted histidine and carboxylate carbons are purple and blue; oxygen and nitrogen retain their standard colors. These are actual retained model coordinates. All views use the same DD1 alignment, camera and scale. The open diamonds in Figure S1 locate these models. They describe geometric cluster representatives, without assigning masked/open or pH-dependent states.

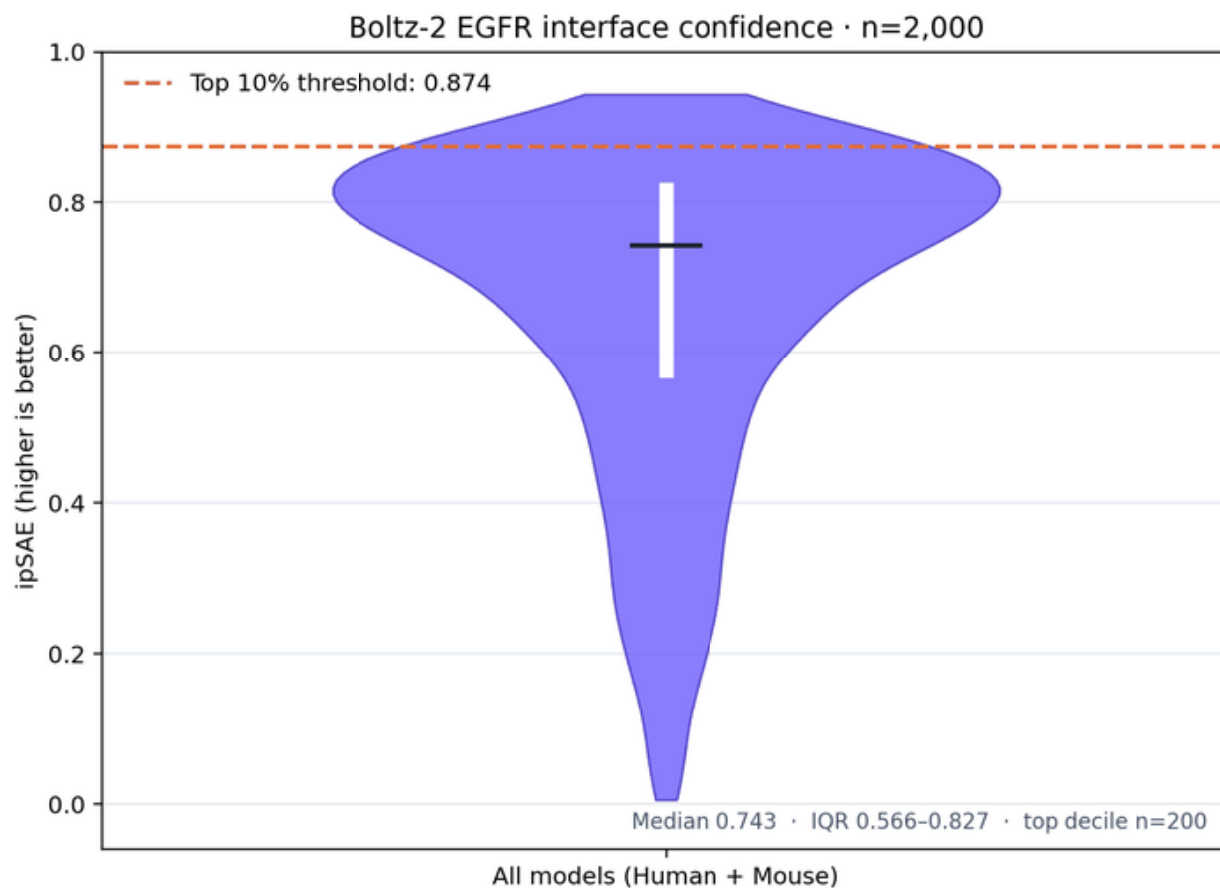


Figure S3: Initial LigandForge design-cohort ipSAE distribution, accessed through LigandAI [5]. The violin pools 2,000 Boltz-2 complex predictions: 1,000 designs, each assessed against human and mouse EGFR D3 [1, 14]. Width indicates the relative density of scores. The white vertical bar spans the middle 50% of values (interquartile range, 0.565585–0.826750); the dark horizontal line marks the median (0.742652). The dashed orange line is the 200th-highest complex score (0.874443), marking the upper 10% of complexes. This descriptive threshold is distinct from the initial selection of ten starting binders, which ranked all species-specific complexes together and retained the ten highest-scoring design identities. The plotted population is the entire initial cohort, not the later 100 selected sequence redesigns. The archived figure is reproduced unchanged.

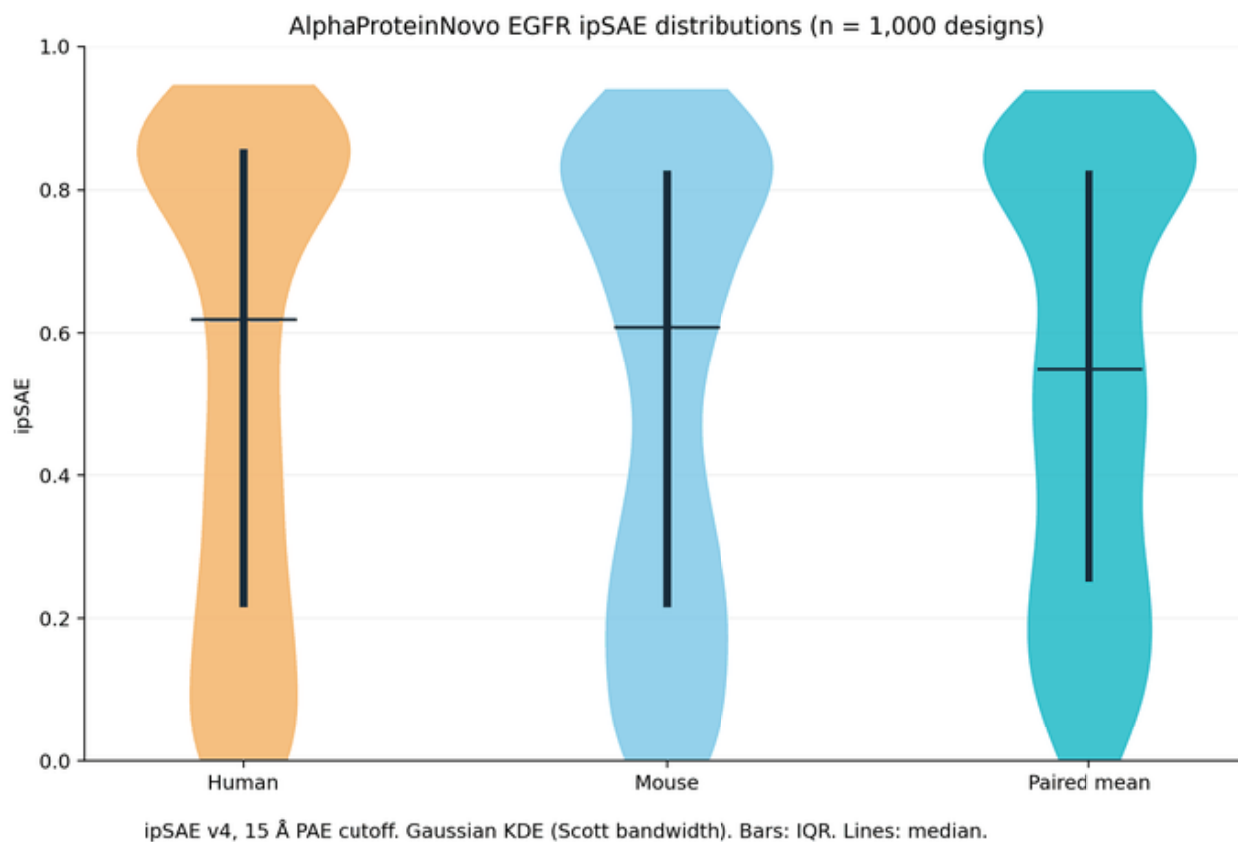


Figure S4: Initial AlphaProtein Novo design-cohort ipSAE distributions [15]. The same 1,000 designs contribute to each violin: one human EGFR D3 score, one mouse score, and the arithmetic mean of that design's two species scores [1, 14]. These are three views of the same cohort. Width indicates the relative density of scores, using the archived Gaussian kernel-density estimate with Scott bandwidth. Dark vertical bars span the middle 50% of values and horizontal lines mark medians: 0.619074 (human), 0.608393 (mouse), and 0.548874 (paired mean). Ten starting binders were selected by paired mean ipSAE; the tenth-ranked value was 0.929295, although this selection cutoff is not drawn. Scores are those of the initial designs before sequence redesign. The archived figure is reproduced unchanged.

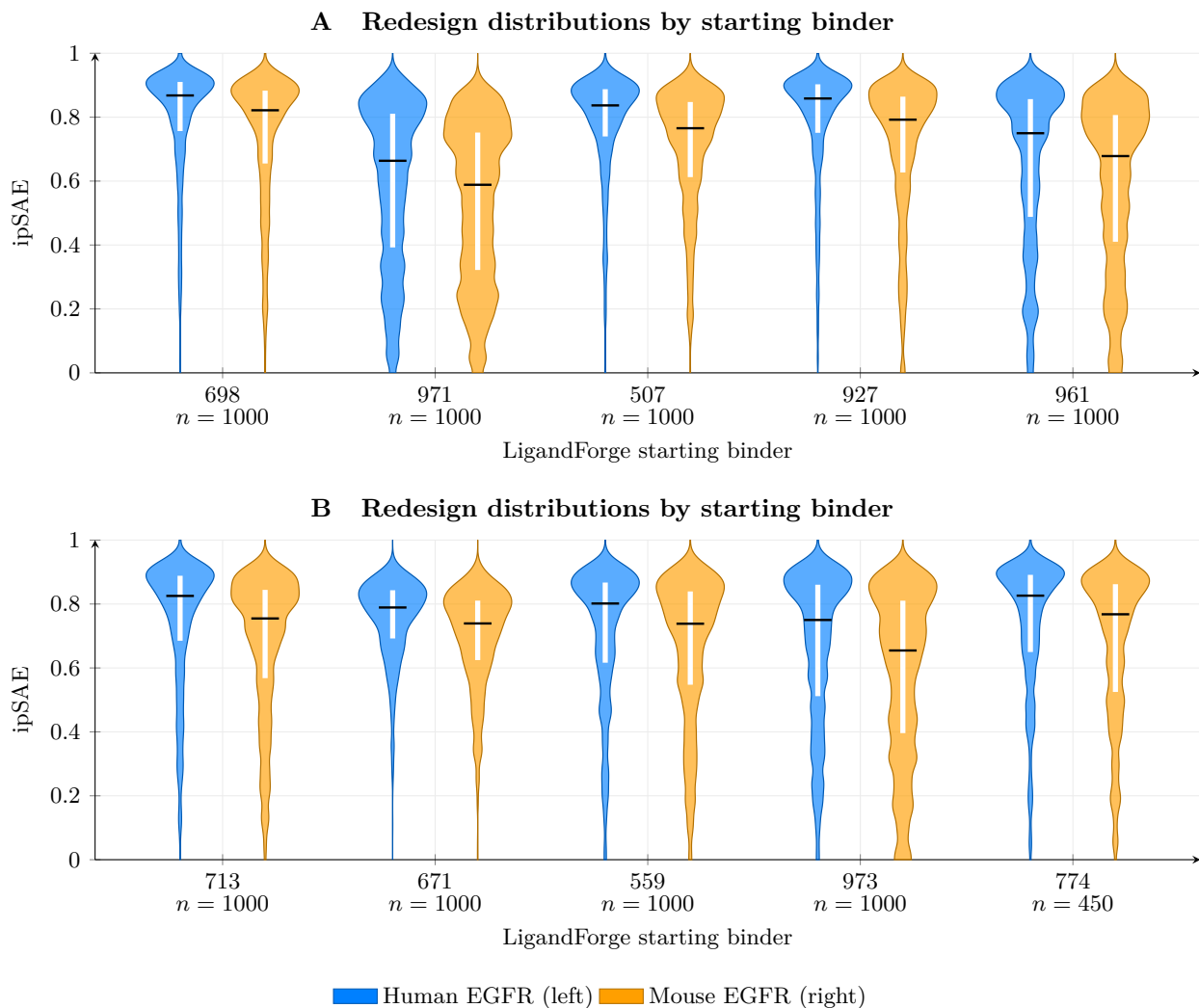


Figure S5: Human and mouse ipSAE distributions for Proton-PottsMPNN redesigns of the ten LigandForge starting binders accessed through LigandAI [12, 13, 5, 6]. Each parent has a pair of violins: human on the left, mouse on the right, giving 20 violins in total. Parent numbers abbreviate the saved identifiers LigandAI_698, _971, _507, _927, _961, _713, _671, _559, _973 and _774, in the displayed order. Each parent generated 1,000 variants. Nine parents have all 2,000 Boltz-2 complex predictions scored; parent 774 has only 450 human and 450 mouse predictions, so its 550 missing pairs are not plotted. The figure contains 18,900 completed model_0 predictions, representing 9,450 matched sequence pairs, before selection of ten variants per parent. Width shows relative score density, normalized separately within each violin; it does not encode sample count. White bars span the middle 50% of scores and black lines mark medians. All violins use Gaussian smoothing with bandwidth 0.025 and reflection at the 0–1 confidence bounds. The same axes and colors apply to both rows. ipSAE uses the larger chain-direction score with 15 Å PAE and distance cutoffs [14]; Boltz-2 predictions have no pH input [1]. Foundry is related-framework attribution for this standalone Proton-PottsMPNN redesign, rather than the execution framework. These distributions show predicted interface confidence, not measured affinity or pH selectivity.

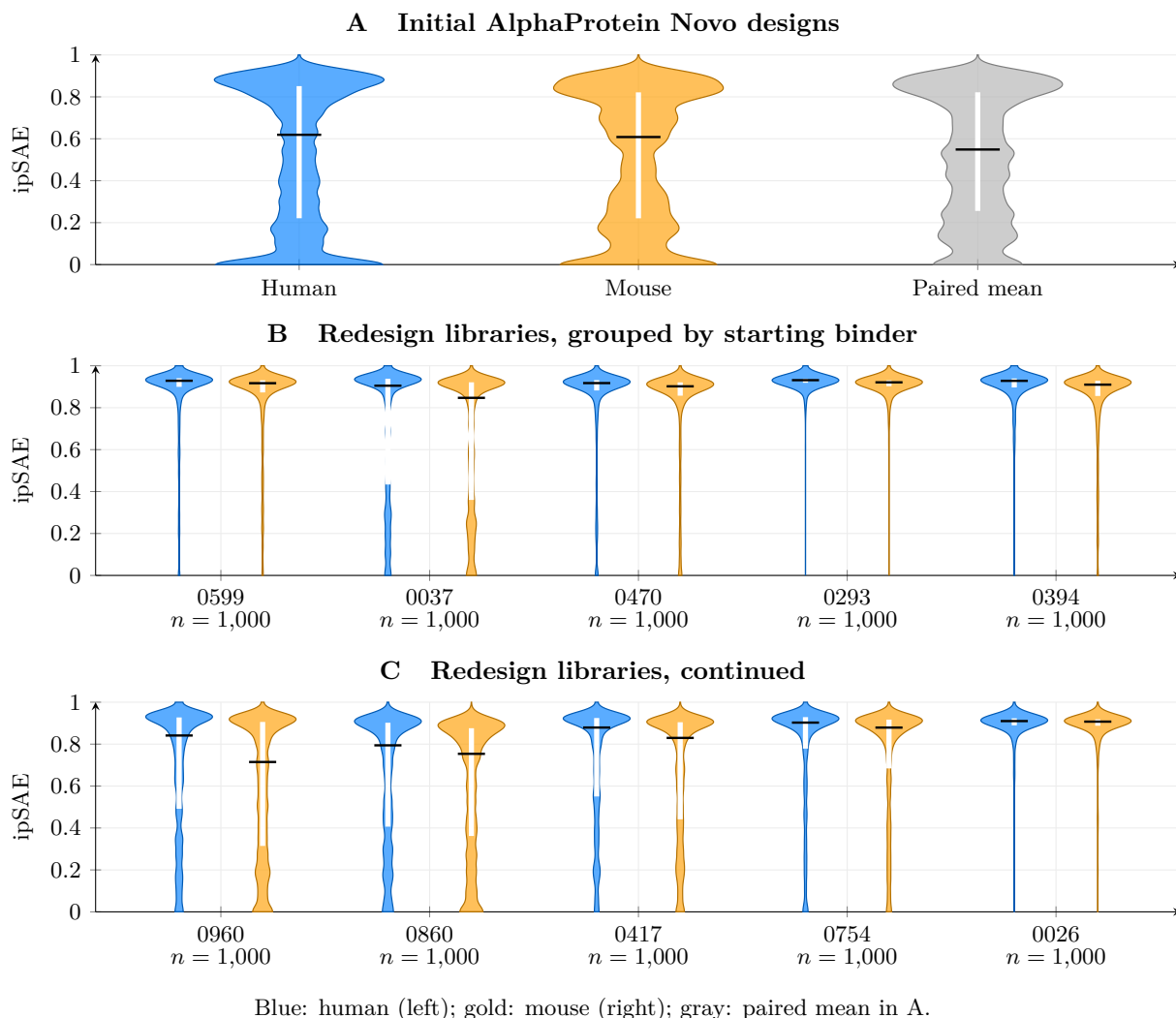


Figure S6: Initial and redesigned AlphaProtein Novo interface-confidence distributions [15, 1, 14]. (A) All 1,000 initial designs, each contributing human and mouse EGFR D3 ipSAE and their arithmetic paired mean. These are three views of the same 1,000 sequences. (B,C) The ten complete Proton-PottsMPNN redesign libraries [12, 13, 6], with 1,000 matched human/mouse pairs per starting binder: 10,000 redesigns and 20,000 complexes before selection. Parent numbers abbreviate APNovo identifiers. Human is left and mouse right within each parent. All ten parent libraries have complete native human/mouse predictions in the verified final job-3835 snapshot. Ten redesigns per displayed parent were selected by descending mean human/mouse ipSAE, yielding 100 selected pairs for subsequent model-based assessments. All violins use Gaussian smoothing with bandwidth 0.025, reflection at 0 and 1, and width normalized separately within each violin; width does not encode count. White bars show interquartile ranges and black lines medians. Scores use the larger chain-direction ipSAE with 15 Å PAE and distance cutoffs. Original scores are reused, with no new structure predictions. Preparation settings differ between initial and redesign campaigns, so this is a descriptive comparison. RosettaCommons Foundry is related-framework attribution for standalone Proton-PottsMPNN. Interface confidence is distinct from affinity, pKa and experimental pH selectivity.

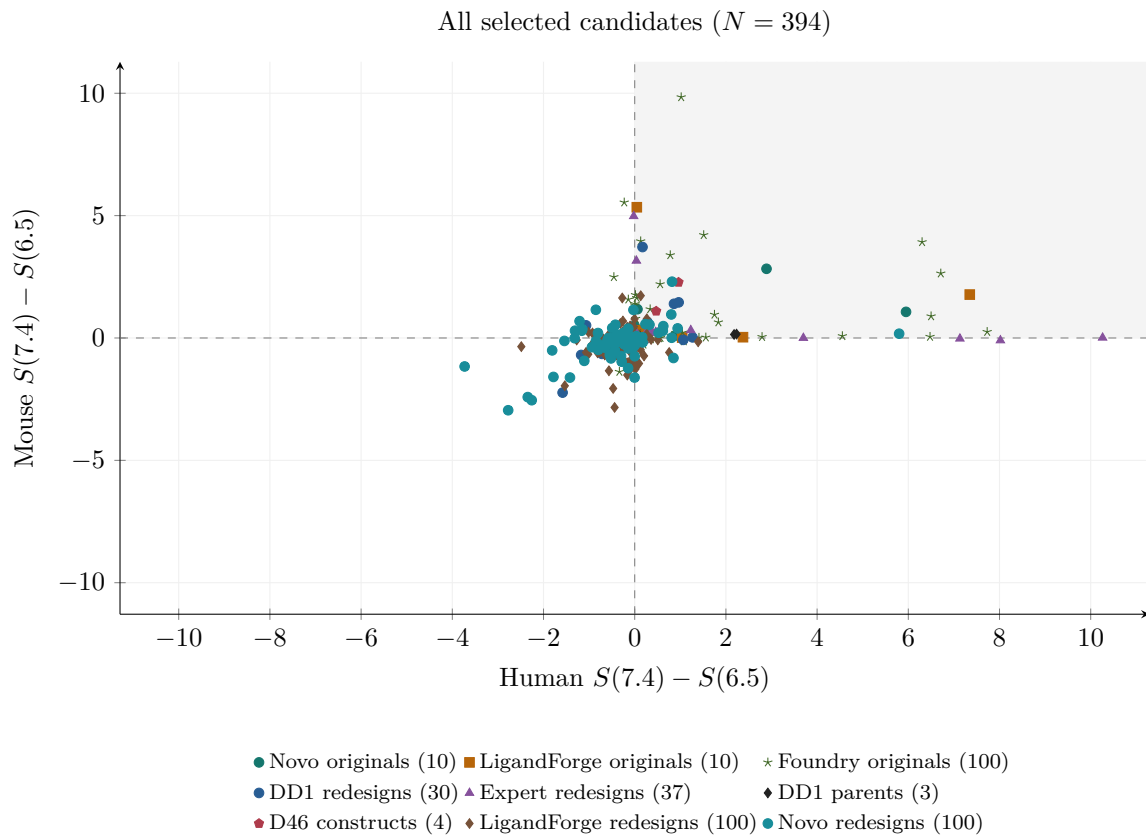


Figure S7: pH-score comparison for all 394 selected candidates, the same inventory used in main Figure 3. Each exact sequence has native human/mouse complexes assessed by standalone Proton-PottsMPNN [12, 13, 6]. Lower scores are favorable; positive $S(7.4) - S(6.5)$ favors pH 6.5. 104 candidates favor acid in both species at nominal binder-site pKas; 96 retain this direction under all common shifts of -1 , 0 and $+1$. All 100 Novo selections are included. Supplementary Data 4 retain both endpoint expectations, exact sequences, sites, protonation patterns and sensitivity flags. These are independent-site model scores, not binding affinities, binding free energies or measured selectivity.

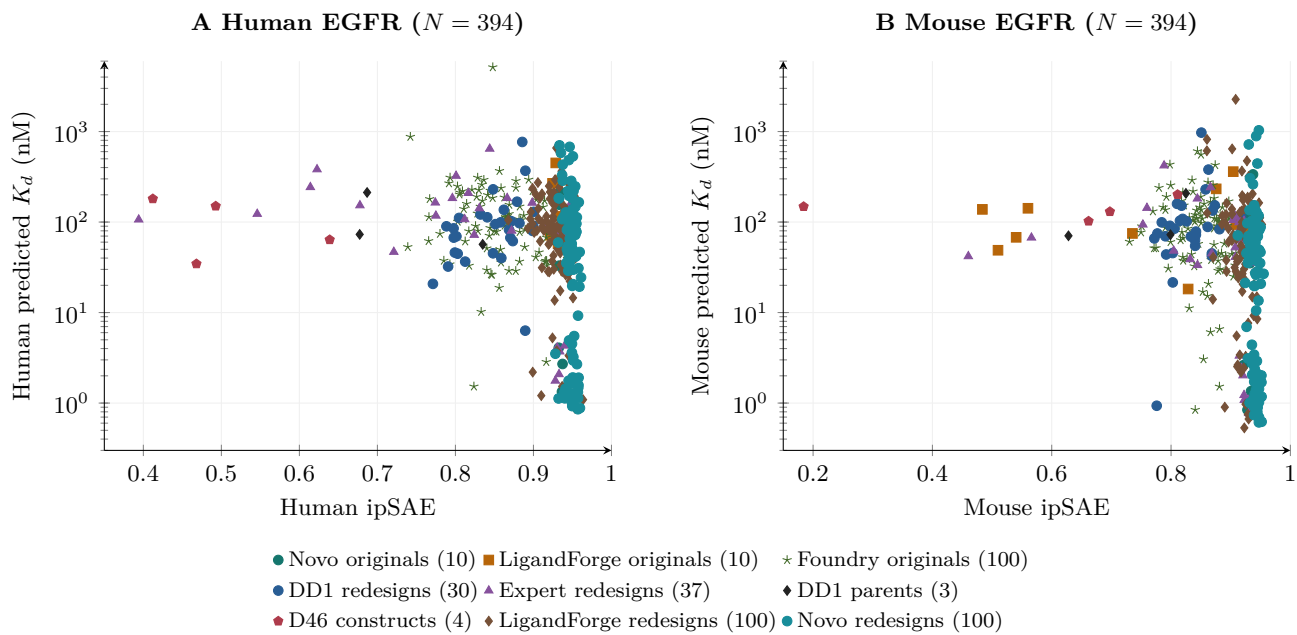


Figure S8: Species-specific predictions underlying main Figure 3C for the same 394 sequences. (A) Human ipSAE versus human predicted K_d ; (B) mouse ipSAE versus mouse predicted K_d . The main panel averages each species pair arithmetically, not geometrically. DeltaForge values [5] have no pH input. The completed Foundry human results are included; all 100 Novo redesigns have completed paired affinity and are included.

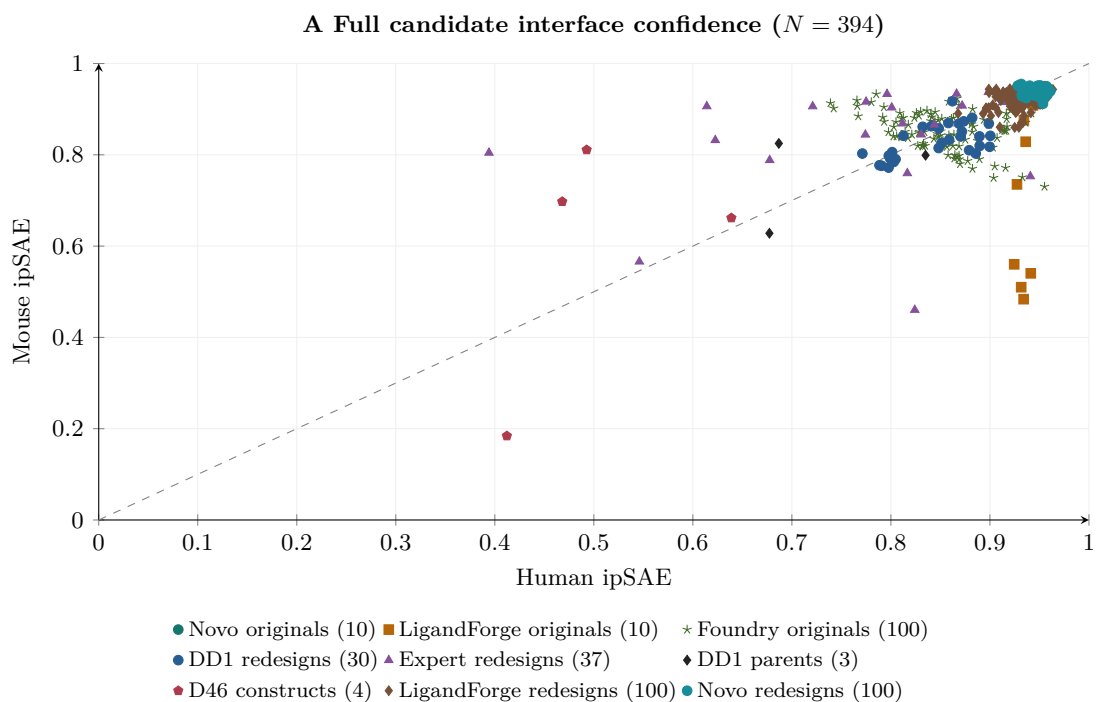


Figure S9: Native human/mouse interface confidence for all 394 exact selected sequences, the same inventory used in all main Figure 3 panels. Original Novo and LigandForge designs, their Proton-PottsMPNN redesigns, Foundry designs, DD1 parents, expert redesigns and D46 constructs retain distinct symbols and colors [16, 12, 13, 15, 5, 6]. All 100 Novo redesigns are included; cohorts retain their different initial selection rules.

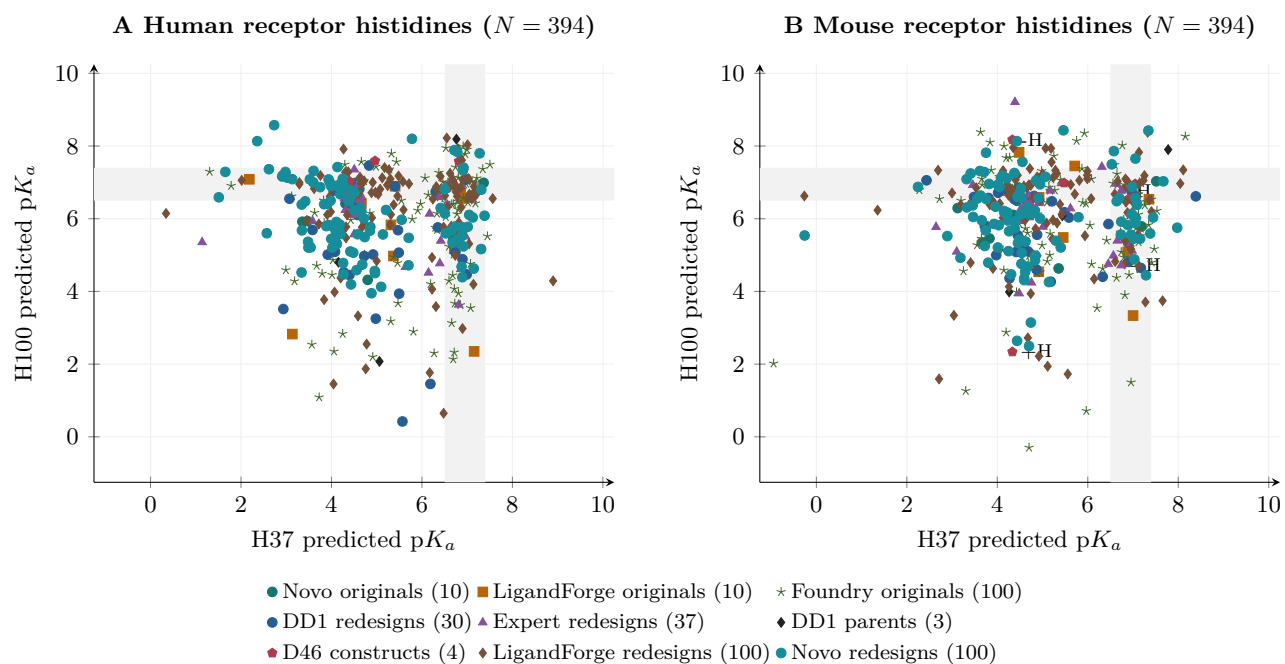
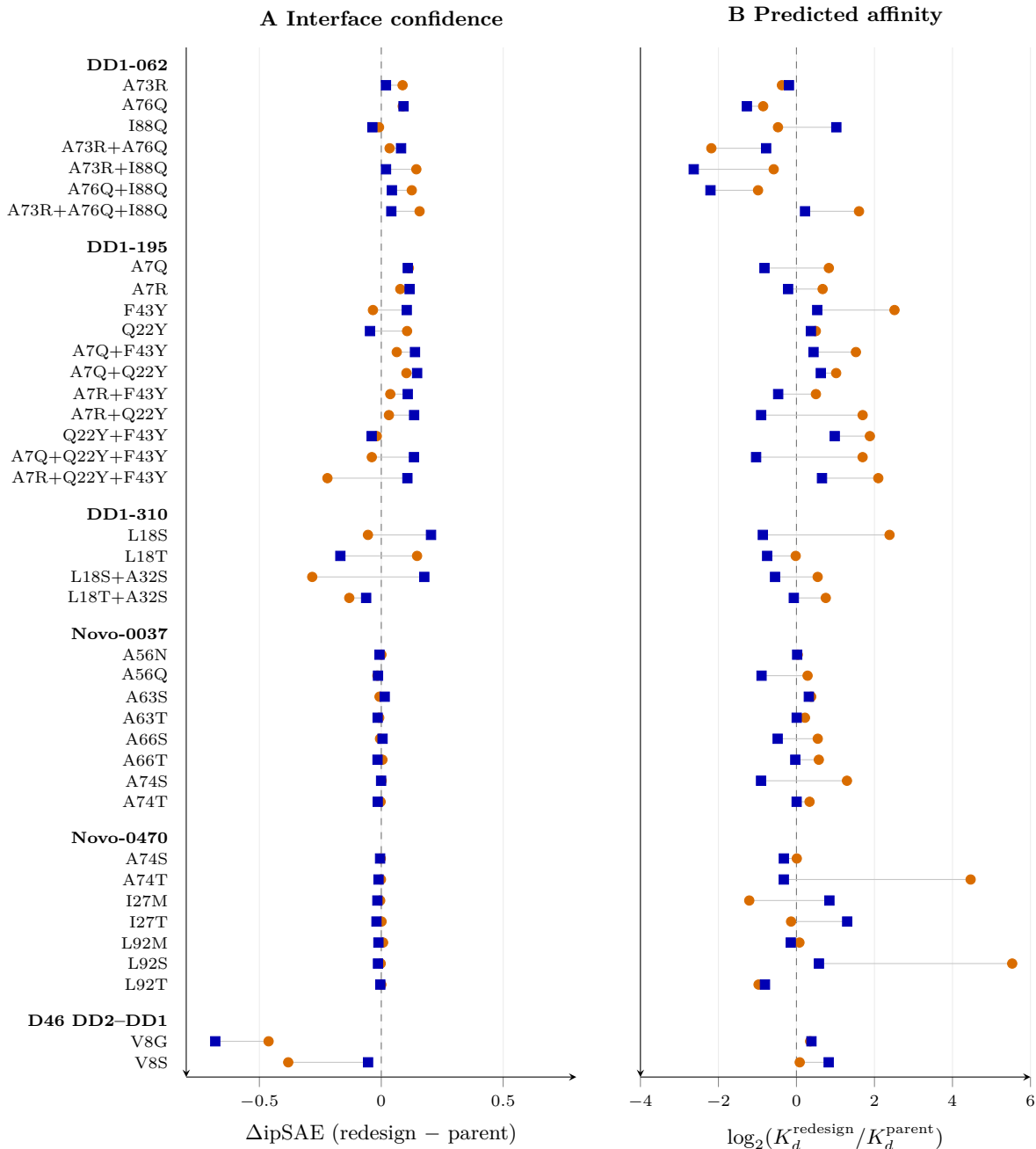


Figure S10: Receptor JustHISpKa [8] predictions for all 394 selected candidate sequences: 394 human unforced pairs and 392 mouse unforced pairs plus two conditional outputs each for DD1-062 A76Q+I88Q and D46 fusion (396 mouse points for 394 sequences). Mouse labels $-H/+H$ identify imposed neighboring binder-acid states as in main Figure 3E; these are separate scenarios, never uncertainty bounds, assay-pH assignments or averages. All 100 Novo redesigns are included. Shaded bands show the inclusive 6.5–7.4 interval; 219 human and 198 unforced mouse pairs have either site in that interval. The ten original LigandForge binders and three DD1 parents use the completed 52-site bound-complex JustHISpKa successor; older PROPKA/receptor-only predictions remain preserved in preceding tables/snapshots. An in-window pKa does not establish pH-dependent affinity.



Orange circles: human EGFR. Blue squares: mouse EGFR. Gray connectors link species for the same redesign.

Figure S11: Expert redesigns compared with their exact-sequence parents ($N = 39$: 37 DD1 redesign variants and two D46 DD2 substitutions). (A) Redesign minus parent ipSAE; positive values indicate increased predicted interface confidence [14]. (B) $\log_2$ of the redesign/parent DeltaForge K_d ratio [5]; negative values indicate lower predicted K_d , and -1 denotes a twofold reduction. Dashed zero lines mark no change. Parent headings define the shared baseline for each group. Values use confidence-selected models independently for each species, never selection by affinity. D46 uses the WT refolded and structure-scored in the same V8 batch; the historical D46 sequence-scored control is not substituted for it. Earlier DD1 parent and redesign preparation/scorer settings differ, so these are descriptive prediction differences rather than isolated mutation effects. Connected dots are paired species predictions, not uncertainty intervals. No explicit pH was used in the affinity prediction. Absolute K_d differences in nM are provided in Table S9; full parent/redesign values, free-energy differences and changes in mouse-minus-human contrasts are retained in the source tables.

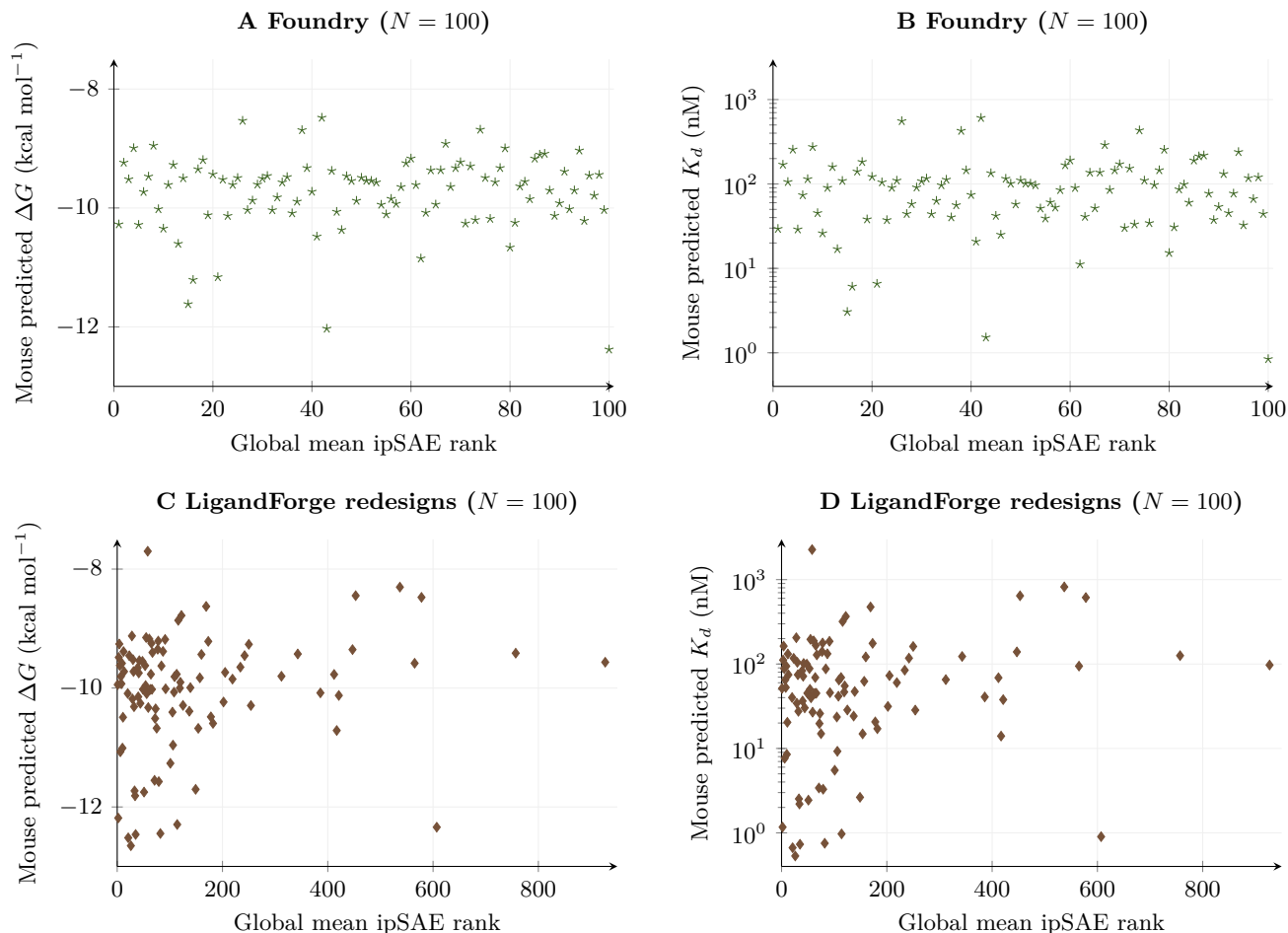


Figure S12: Mouse EGFR affinity predictions versus global rank by arithmetic mean human/mouse ipSAE. (A,B) The 100 original Foundry-lineage RFdiffusion3 designs [16, 6]; (C,D) the 100 selected LigandForge-derived standalone Proton-PottsMPNN redesigns [12, 13, 5, 6]. Left panels show DeltaForge predicted ΔG ; right panels show its returned calibrated K_d on logarithmic nM axes [5]. Ranks are assigned within each campaign: Foundry retains ranks 1–100 among 10,742 paired designs. LigandForge retains ten redesigns per starting binder, with global ranks 1–927 among 9,450 completed pairs pooled across all ten starting binders; the 550 incomplete pairs are excluded from ranking. The latter selection is not the global top 100. Horizontal ranges differ between rows, while corresponding vertical scales match. Lower rank indicates higher mean interface confidence; more negative predicted ΔG and lower predicted K_d favor binding. No affinity prediction used a pH input. The exact sequence identities, ranks and served values are retained in the accompanying source-linked CSV; no new scoring or sequence selection was performed.

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